API Specification Document

DTV – CCS2

Author(s):

Revision History

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	3.1.7 Direct Channel Select 3.1.8 Dtv Initialize 3.2.1 Set Auto Tune Mode 3.2.2 Set Auto Broadcast Search Mode 3.2.1 Set Auto Event Relay Mode 3.2.1 Set Closed Caption Mode 3.2.1 Set Audio Mode 3.1.9 Reception Monitor (Test Mode) 3.2.6 Set Physical Channel 3.3.3 Get Version Info 3.3.4 Get Program Info 3.4.7 Reception Monitor Update (Test Mode) 3.1.10 Dtv Firmware Update 3.4.6 Dtv Status Update 3.4.8 Dtv Firmware Update 3.3.5 Get RMP Manufacturer Id	
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1 Purpose of this Document

1.1 Introduction to the document

This document is used to provide information related to the APIs provided by DTV Service in CCS2 project.
The document can be used by:

- Feature Owner to verify all the features offered by DTV Service
- App Developers to understand how to use the APIs to interface and available features offered by DTV Service

1.2 Scope

The scope of this document is to focus on the APIs offered by DTV Service. It also takes into account the various pre-conditions, usage and results of the APIs

1.3 Context

This is a preliminary draft version of the document. Document will be updated for below points during next Engineering Cycle documentation phase.

- Review comments from Feature Owner, Architect & Specialist
- Robustness (Error Handling, Low Voltage Handling, Configuration Handling)
- Size & Performance Criteria

2 Service Information

Service Name: com.bosch.dtv.service

Package : com.bosch.dtv.service

Persmission : com.bosch.permission.DTV

3 API

This section describes the APIs provided by DTV Service

3.1 DTV Commands

3.1.1 Auto Preset

Feature	**Auto Preset for TV1/TV2**		
Program Epic	CCS2-3554		
API Method Name	***dtvAutoPreset***		
Parameters	**DtvAutoPresetRequest**	***request***	*START_HOME_AREA_UPDATE* *START_TRAVEL_AREA_UPDATE* *STOP*
	IDtvMethodCallback	***callback***	
Return	**void**		
Result	**IDtvMethodCallback:: onDtvAutoPresetResult** Refer Dtv Result table for information on result		
Events to Follow	**(optional) IDtvEventsCallback::onDtvAutoPresetUpdate** Refer Auto Preset Update		
	(optional) IDtvEventsCallback:: onDtvPresetListUpdate Refer Preset List Update		

Feature Description:

User can perform Auto Preset to scan entire band and store receivable channels into TV1 or TV2 preset list. Up to 12 stations can be stored into each preset list.

- Result would be provided through ***onDtvAutoPresetResult.***
- When Auto Preset is started successfully, Intermediatory and concluding events would be provided, through ***onDtvAutoPresetUpdate***, giving information related to each physical channels from 13 to 52 scanned and reports if any station was found or not.
- During Auto Preset, if at least one station was found, Preset list would be updated and provided through ***onDtvPresetListUpdate*** and the first available stored preset channel (smallest preset number) in that preset list would be tuned.
- If no station was found or cancelled by user, then TV module will retune the formerly tuned last channel, if any was tuned before. Last preset list stays unchanged.

Usage:

- ***dtvAutoPreset*** with *START_HOME_AREA_UPDATE* can be requested when current TV type is TV1, and user selects Auto Preset option from Settings - ***dtvAutoPreset*** with *START_TRAVEL_AREA_UPDATE* can be requested when current TV type is TV2, and user selects Auto Preset option from Settings - ***dtvAutoPreset*** with *STOP* can be requested when user selects Cancel from Auto Preset Update popup
Pre-Condition:
• TV Module is running, and DTV service is available. There is no automatic station search or seek running.
Important:
• During Auto Preset, other requests (except Auto Preset Stop) cannot be handled and would result in error response. • To ensure user does not perform other operations while Auto Preset is ongoing, Popup can be displayed
Sequence:
For Complete Feature Sequence, refer Auto Preset

3.1.2 Channel Seek

Feature	**Channel Seek Up/Down**		
Program Epic	CCS2-1617		
API Method Name	***dtvChannelSeek***		
Parameters	**DtvSeekOperation**	***operation***	*UP* *DOWN* *STOP*
	IDtvMethodCallback	***callback***	
Return	**void**		
Result	**IDtvMethodCallback:: onDtvChannelSeekResult** Refer Dtv Result table for information on result		
Events to Follow	**(optional) IDtvEventsCallback:: onDtvChannelSeekUpdate** Refer Channel Seek Update		

Feature Description:

User can perform Channel Seek to search the band, to find other receivable channels.

- Result would be provided through ***onDtvChannelSeekResult.***
- When Channel Seek is started successfully, Intermediatory and concluding events would be provided, through ***onDtvChannelSeekUpdate***, giving information related to different physical channels searched and reports if any channel was found
- If no channel was found or seek Stopped by user, then TV module will retune the formerly tuned last channel, if any was tuned before.
- If a channel is found during seek, current seek would be concluded and the found channel would be tuned and played.
- Operation *Up* will search band in increasing order from the next physical channel of currently tuned physical channel until the one channel before current channel. E.g., when current channel is 36 then from 37 to 52 then 13 to 35.
- Operation *Down* will search band in decreasing order opposite to Seek up.

Pre-Condition:

- TV Module is running, and DTV service is available. There is no automatic station search or Auto preset in progress.

Important:

- During Seek, other requests (except Seek) cannot be handled and would result in error response.
- To ensure user does not perform other operations while seek is ongoing, Popup can be displayed.
- TV module has some limitations due to which the Seek should be Aborted\Cancelled in a few Use Cases before starting another operation. The below listed a few possible Use Cases where App should first Stop the Seek in Progress before triggering another Request.

Notes:
Implicit requirements needed due to tv module limitation:
Precondition: TV1 and TV2 both have presets stored

Use case 1:
User in TV1, perform seek
Change source to TV2
Requirement: TV2 last played preset type should be tuned
Implicit handling: Seek has to be aborted and then tv2 last preset tune request should be sent.

Use case 2:
User in TV1, perform seek
Press HK SWC Next / Prev (short)
Requirement: ?
Implicit handling : seek has to be aborted and then channel up /down request need to be sent

Use case 3:
User in TV1, perform seek Select preset.
Requirement: ?
Implicit handling: seek has to be aborted and then preset select request need to be sent.

Use case 4:
User in TV1, perform seek.
Change Audio Source from TV1 to AM\FM and quickly changes it back to TV1.
Requirement: TV1 last played preset type should be tuned
Implicit handling: seek has to be aborted when TV is exited, and Source changed to AM\FM.

Sequence:

For Complete Feature Sequence, refer Channel Seek

3.1.3 Channel Up Down

Feature	Channel Up/Down (Next/Previous)

Program Epic	CCS2-4631		
API Method Name	***dtvChannelUpDown***		
Parameters	**DtvChannelUpDownOperation**	***operation***	*UP* *DOWN*
	IDtvMethodCallback	***callback***	
Return	**void**		
Result	**IDtvMethodCallback:: onDtvChannelUpDownResult** Refer Dtv Result table for information on result		

Feature Description:

User can perform channel up / down operation to tune to the next / previous presets from the preset list

- Result would be provided through ***onDtvChannelUpDownResult.***
- The ***dtvChannelUpDown*** request performs the requested operation on **current preset type** only.
- Different actions that can be performed with **DtvChannelUpDownOperation** parameter
  - *UP*: Tune to the next sub service of current preset if available, else tune to next preset
  - *DOWN:* Tune to the previous sub service of current preset if available, else tune to previous preset

Example:

Preset 1: 011, 012(unique service), 013(unique service)
Preset 2: 021(no unique service in sub channels)
Preset 3: Empty
Preset 4: 041, 043(unique service)
Preset 5: 051, 052(unique service)
Preset 6 – 12: Empty

Current Tuned Preset: Preset 1, Channel Number: 011

UP: 012 -> 013 -> 021 -> 041 ->043 -> 051->052->011

DOWN: 052->051->043->041->021->013->012->011

If Preset list is empty or if preset list has no other unique service to Tune to, ***dtvChannelUpDown*** request will fail and provide Result as NO_SERVICE.

Pre-Condition:

• TV Module is running, and DTV service is available. There is no automatic station search or Auto preset in progress. • Program Guide or Program Info are not open.
Sequence:
For Complete Feature Sequence, refer Channel Up Down

3.1.4 Select Preset Channel

Feature	Preset Channel Select/Store for TV1/TV2		
Program Epic	CCS2-3556		
API Method Name	***dtvSelectPresetChannel***		
Parameters	**DtvPresetType**	***presetType***	*HOME* *DRIVE*
	int	***presetNumber***	*01-12*
	IDtvMethodCallback	***callback***	
Return	**void**		
Result	**IDtvMethodCallback:: onDtvSelectPresetChannelResult** Refer Dtv Result table for information on result		

Feature Description:

User can select any preset from the preset list, to tune.

- Result would be provided through ***onDtvSelectPresetChannelResult.***
- Any ***dtvSelectPresetChannel*** request for any empty preset would lead to error result.
- If ***result*** is OK,
  - requested ***presetNumber*** would be tuned

Pre-Condition:

- TV Module is running, and DTV service is available. There is no automatic station search or Auto preset in progress.

Sequence:

For Complete Feature Sequence, refer Select Preset Channel

3.1.5 Store Preset Channel

Feature	Preset Channel Select/Store for TV1/TV2		
Program Epic	CCS2-3556		
API Method Name	***dtvStorePresetChannel***		
Parameters	**DtvPresetType**	***presetType***	*HOME* *DRIVE*
	int	***presetNumber***	*01-12*
	IDtvMethodCallback	***callback***	
Return	**void**		
Result	**IDtvMethodCallback:: onDtvStorePresetChannelResult** Refer Dtv Result table for information on result		
Events to Follow	**(optional) IDtvEventsCallback:: *onDtvPresetListUpdate*** Refer Preset List Update		

Feature Description:

User can store the currently playing station in any preset

- Result would be provided through ***onDtvStorePresetChannelResult.***
- If the ***presetNumber*** requested to store, already has preset information, it would be overwritten with the current playing station information.
- If ***result*** is OK,
  - Requested ***presetNumber*** would be stored in preset list and tuned
  - Updated preset list would be provided through ***onDtvPresetListUpdate***

Pre-Condition:

- TV Module is running, and DTV service is available. There is no automatic station search or Auto preset in progress.

Sequence:

For Complete Feature Sequence, refer Store Preset Channel

3.1.6 Display Banner

Feature	Display Banner (overlaid on TV video)		
Program Epic	CCS2-3559		
API Method Name	***dtvDisplayBanner***		
Parameters	**DtvBannerOperation**	***operation***	*SHOW* *HIDE*
	int	***duration***	*0* *1-255*
	IDtvMethodCallback	***callback***	
Return	**void**		
Result	**IDtvMethodCallback:: onDtvDisplayBannerResult** Refer Dtv Result table for information on result		

Feature Description:

Application can request to show/ hide banner, which shows the current tuned station's program details.

- Result would be provided through ***onDtvDisplayBannerResult.***
- If operation *SHOW* is requested,
  - If *duration* is 0, Banner would be shown till *HIDE* is requested.
  - If *duration* is 1-255, Banner would be shown for that many seconds and hidden automatically.
- If operation *HIDE* is requested, Banner would be hidden, if being shown.
- If banner is being shown and a new ***dtvDisplayBanner*** request is made, the timer would be restarted.
- Banner would be automatically shown whenever a channel is Tuned.

Pre-Condition:

- TV Module is running, and DTV service is available.

Sequence:

For Complete Feature Sequence, refer Display Banner

3.1.7 Direct Channel Select

Feature	**Direct Channel Input**		
Program Epic	CCS2-5148		
API Method Name	***dtvDirectChannelSelect***		
Parameters	**int**	***threeDigitNumber***	*000~199 (Full seg)* *600~799 (One seg)*
	int	***branchId***	*0-9*
	IDtvMethodCallback	***callback***	
Return	**void**		
Result	**IDtvMethodCallback:: onDtvDirectChannelSelectResult** Refer Dtv Result table for information on result		

Feature Description:

User can request to tune to a channel using its threeDigitNumber and its branchId.

- By default, *branchId* should be requested as 1 since branch id would not be stored by application.
- Result would be provided through onDtvDirectChannelSelectResult.
- If the requested threeDigitNumber – branchId is not stored in current preset list or not the current tuned non-stored preset channel, result would be provided as NO_SERVICE.

Pre-Condition:

- TV Module is running, and DTV service is available.

Sequence:

For Complete Feature Sequence, refer Direct Channel Select

3.1.8 Dtv Initialize

Feature	**TV Settings: Initialization (TV1 / TV2 / Settings / Factory Reset)**		
Program Epic	CCS2-3564		
API Method Name	***dtvInitialize***		
Parameters	**DtvInitializeOperation**	***operation***	*ALL_DATA* *CONFIGURED_SETTINGS* *HOME_PRESET* *DRIVE_PRESET*
	IDtvMethodCallback	***callback***	
Return	**void**		
Result	**IDtvMethodCallback:: onDtvInitializeResult** Refer Dtv Result table for information on result		
Feature Description:			

User can initialize either Home or Drive or Settings or All Dtv Data

- Result would be provided through ***onDtvInitializeResult.***
- Refer below table for the data which would be initialized based on requested ***operation***

Data	Initial Value	Operation
Audio Mode	MAIN	*ALL_DATA or SETTINGS*
Closed Caption Mode	OFF	*ALL_DATA or SETTINGS*
Auto Tune Mode	AUTO	*ALL_DATA or SETTINGS*
Auto Broadcast Search Mode	RELAY_AFFILIATED_STATION_SEARCH	*ALL_DATA or SETTINGS*
Auto Event Relay	ON	*ALL_DATA or SETTINGS*
Preset List (Home Preset)	Empty	*ALL_DATA or HOME_PRESET*
Region ID (Home Preset)	255	*ALL_DATA or HOME_PRESET*
Zip code (Home Preset)	0	*ALL_DATA or HOME_PRESET*
Preset List (Drive Preset)	Empty	*ALL_DATA or DRIVE_PRESET*
Region ID (Drive Preset)	255	*ALL_DATA or DRIVE_PRESET*
Zip code (Drive Preset)	0	*ALL_DATA or DRIVE_PRESET*

- If current playing channel is from Home Preset and user requests for ***dtvInitialize*** with *operation as HOME_PRESET* or *ALL_DATA*, the current playback would be stopped, and last channel information would be cleared. Same logic is applicable for vice-versa.
- If current playing channel is from Home Preset and user requests for ***dtvInitialize*** with *operation as DRIVE_PRESET*, the current playback would be resumed after initialization. Same logic is applicable for vice-versa.
- Current playing channel or last channel information would not be cleared in case of ***dtvInitialize*** with *operation as SETTINGS.*
- *If application is storing local data of any of the below types, it must be cleared on receiving the* **onDtvInitializeResult** as **OK**

DtvInitialize Operation	TV1 last played preset number	TV2 last played preset number	TV1 current program info	TV2 current program info	Home Area zipcode & prefecture	Travel area zipcode & prefecture	Audio Mode, Closed caption mode, Auto tune mode, Auto Broadcast Search mode, Auto Event relay
TV1	x		x		x		
TV2		x		x		x	
Settings							x
ALL	x	x	x	x	x	x	x

Pre-Condition:

• TV Module is running, and DTV service is available.
Sequence:
For Complete Feature Sequence, refer Dtv Initailize

3.1.9 Reception Monitor (Test Mode)

Feature	**Developer Test Mode (ENG Menu)**		
Program Epic	CCS2-3575		
API Method Name	***dtvReceptionMonitorSession***		
Parameters	**DtvReceptionMonitorState**	***state***	*STOP,* *START*
	IDtvMethodCallback	***callback***	
Return	**void**		
Result	**IDtvMethodCallback:: onDtvReceptionMonitorSessionResult** Refer Dtv Result table for information on result		
Events to Follow	**(optional) IDtvEventsCallback:: *onDtvReceptionMonitorUpdate*** Refer Reception Monitor Update (Test Mode)		
Feature Description:			
Reception Monitoring can be started/stopped using this interface • Result would be provided through ***onDtvReceptionMonitorSessionResult.*** • On request of state as *START*, ***onDtvReceptionMonitorUpdate*** would be provided every second and would continue till STOP state is requested.			
Pre-Condition:			
• TV Module is running, and DTV service is available.			
Sequence:			
For Complete Feature Sequence, refer Reception Monitor (Test Mode)			

3.1.10 Dtv Firmware Update

Feature	Developer Test Mode (ENG Menu)		
Program Epic	CCS2-3575		
API Method Name	***dtvPerformFirmwareUpdate***		
Parameters	IDtvMethodCallback	*callback*	
Return	void		
Result	**IDtvMethodCallback:: onDtvPerformFirmwareUpdateResult** Refer Dtv Result table for information on result		
Events to Follow	**(optional) IDtvEventsCallback:: onDtvFirmwareUpdate** Refer		

Feature Description:

Firmware update of Dtv Module can be initiated using this interface

- Result would be provided through ***onDtvPerformFirmwareUpdateResult***
- Progress of firmware update would be provided through ***onDtvFirmwareUpdate***

Pre-Condition:

- TV Module is running, and DTV service is available.

Sequence:

For Complete Feature Sequence, refer Dtv Firmware Update

3.1.11 Display EPG

Feature	EPG (8 Days Program Schedule for multiple channels), Program Info (Detailed EPG)		
Program Epic	CCS2-3552, CCS2-3553		
API Method Name	***dtvDisplayEpg***		
Parameters	DtvEpgOperation	*operation*	
	IDtvMethodCallback	*callback*	
Return	void		

Result	**IDtvMethodCallback:: onDtvDisplayEPGResult**		
	DtvResult	***result***	Refer Dtv Result table for information on result
	DtvEpgScreenState	***screenState***	
	DtvEpgButtons	***buttonState***	
Events to Follow	**(optional) IDtvEventsCallback:: onDtvEpgUpdate** Refer Dtv EPG Update		
Feature Description:			

EPG Program Guide and Program Information can be accessed and operated using this interface

- Result would be provided through **onDtvDisplayEPGResult**

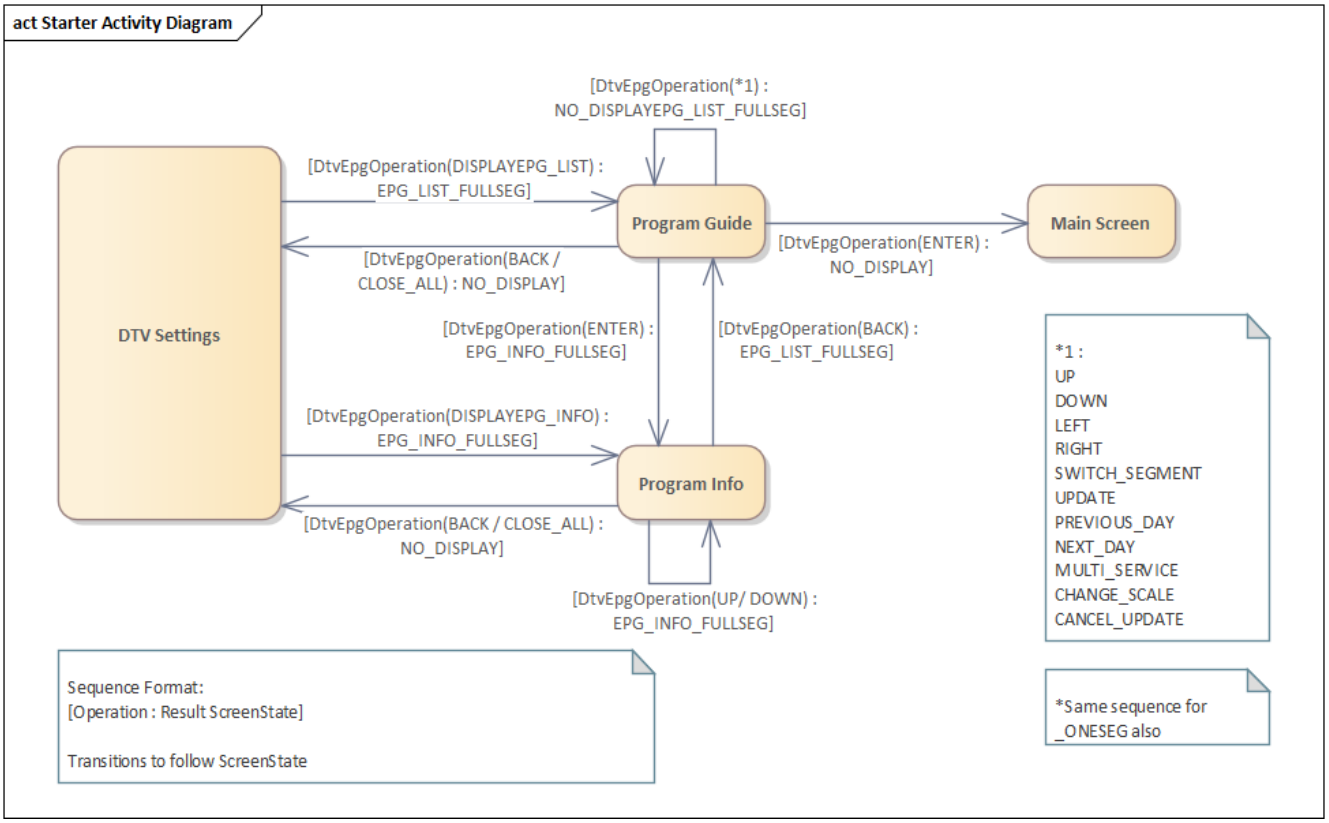
DtvEpgOperation	operation
DISPLAYEPG_LIST	To open EPG List
DISPLAYEPG_INFORMATION	To open EPG Information
UP	To scroll Up
DOWN	To scroll Down
LEFT	To scroll Left
RIGHT	To scroll Right
BACK	To go back to previous screen
ENTER	Refer detailed description below
SWITCH_SEGMENT	To switch between program information for 1seg and 12seg
UPDATE	To manually update EPG
PREVIOUS_DAY	To display program information of previous day
NEXT_DAY	To display program information of next day
MULTI_SERVICE	To display program information of sub channels
CHANGE_SCALE	To adjust the zoom level of EPG Table
CANCEL_UPDATE	To cancel the manual EPG update
CLOSE_ALL	Refer detailed description below

- *DtvEpgScreenState* describes the current state of EPG

DtvEpgScreenState	description
NO_DISPLAY	EPG List or EPG Information is not open (TV Video shown)
EPG_LIST_FULLSEG	EPG List shown with 12 seg program information
EPG_INFO_FULLSEG	EPG Info shown with 12 seg program information
EPG_LIST_ONESEG	EPG List shown with 1 seg program information
EPG_INFO_ONESEG	EPG Info shown with 1 seg program information

- All screen transitions should follow the screen state
- Transit to Program Guide screen after receiving ***DtvEpgScreenState*** as EPG_LIST_FULLSEG / EPG_LIST_ONESEG result for the ***dtvDisplayEpg*** *DISPLAYEPG_LIST* request
- Transit to Program Info screen after receiving ***DtvEpgScreenState*** EPG_INFO_FULLSEG / EPG_INFO_ONESEG result for the ***dtvDisplayEpg*** *DISPLAYEPG_INFO* request
- If ***DtvEpgScreenState*** is NO_DISPLAY, EPG session is closed
  - Transit to main screen if NO_DISPLAY received as result for ENTER ***DtvEpgOperation***
  - Transit to previous screen if NO_DISPLAY received as a result for other ***DtvEpgOperation***
- When **DtvEpgOperation** UPDATE is called, intermediatory and conclusive updates would be provided through ***onDtvEpgUpdate***
- **DtvEpgButtons** received through result inform if the operation is allowed.

DtvEpgButtons	description
isValid_up	true: valid. Operation allowed. Button can be enabled false: invalid. Operation now allowed. Button can be disabled
isValid_down	
isValid_left	
isValid_right	
isValid_back	
isValid_enter	
isValid_switchSegment	
isValid_update	
isValid_previousDay	
isValid_nextDay	
isValid_multiService	
isValid_changeScale	



Pre-Condition:

- TV Module is running, and DTV service is available.

Important:

• Tuning / Search / Seek / Auto Preset operations are not allowed when EPG is open. Any request would result in failure.
Sequence:
For Complete Feature Sequence, refer EPG

3.1.12 Operate BML

Feature	CCS2-3562		
Program Epic	**BML Browser**		
API Method Name	***dtvOperateBml***		
Parameters	**DtvBmlButton**	***button***	
	DtvBmlButtonState	***state***	*PRESS* *RELEASE*
Return	**void**		
Result	**void**		
Feature Description:			

User can operate BML using this interface

- There is no result provided for this interface
- Possible operations with BML which can be mapped to various buttons on remote / ‘d’ button

DtvBmlButton
NUM_1
NUM_2
NUM_3
NUM_4
NUM_5
NUM_6
NUM_7
NUM_8
NUM_9
NUM_10
UP
DOWN
LEFT
RIGHT
ENTER
BACK
DATA_BROADCAST
BLUE
RED
GREEN
YELLOW

- To start data broadcasting (on press of ‘d’ button), ***dtvOperateBml*** with *DATA_BROADCAST* should be used
- The successful opening of BML Data broadcast session can only be judged by the beep sound heard
- ***dtvOperateBml*** with *DATA_BROADCAST* would also toggle between opening BML and closing of BML.
- To close BML, ***dtvForcedExitBml*** should be used to guarantee the closure of BML Data broadcast session
- On press action of the button, application should use ***dtvOperateBml*** with relevant ***button*** and ***state*** as *PRESS*
- On (short/long) release action of the button, ***dtvOperateBml*** with relevant ***button*** and ***state*** as *RELEASE* should be used

Pre-Condition:

• TV Module is running, and DTV service is available.
Sequence:
For Complete Feature Sequence, refer BML Data Broadcast

3.1.13 Forced Exit BML

Feature	**BML Browser**		
Program Epic	CCS2-3562		
API Method Name	***dtvForcedExitBml***		
Parameters	**IDtvMethodCallback**	***callback***	
Return	**void**		
Result	**IDtvMethodCallback:: onDtvForcedExitBmlResult** Refer Dtv Result table for information on result		

Feature Description:
Application can use this interface to ensured guaranteed closure of BML Data broadcast session • Result would be provided through ***onDtvForcedExitBmlResult*** • If BML Data broadcast session is open and ***dtvForcedExitBml*** sent, BML session would be closed, and SUCCESS result would be given • If BML Data broadcast session is not open and ***dtvForcedExitBml*** sent, no action and SUCCESS result would be givenl • This interface can be used on exit of BML screen or on back button press in the BML screen (not the back button in remote)
Pre-Condition:
• TV Module is running, and DTV service is available.
Sequence:
For Complete Feature Sequence, refer BML Data Broadcast

3.1.14 Broadcast Search

Feature	**Manual Foreground Broadcast Search (Relay/Affiliated Station) / Automatic Foreground Broadcast Search (Relay/Affiliated Station)**		
Program Epic	CCS2-3560 / CCS2-3561		
API Method Name	***dtvBroadcastSearch***		
Parameters	**DtvBroadcastSearchMode**	***mode***	*START_RELAY_SEARCH* *START_AFFILIATED_SEARCH* *START_RELAY_AFFILIATED_SEARCH* *STOP_SEARCH*
	IDtvMethodCallback	***callback***	
Return	**void**		
Result	**IDtvMethodCallback:: onDtvBroadcastSearchResult** Refer Dtv Result table for information on result		
Events to Follow	**(optional) IDtvEventsCallback:: onDtvEventUpdate** Refer Dtv Event Update		
	(optional) IDtvEventsCallback:: onDtvBroadcastSearchUpdate Refer Broadcast Search Update		
Feature Description:			

User can perform Broadcast search to

- Find affiliate / relay station / relay and affiliate / stop broadcast search (manual)
- stop broadcast search (automatic)

- Result would be provided through **onDtvBroadcastSearchResult.**
- When Broadcast search (Manual – started by using ***dtvBroadcastSearch***) is started, **onDtvEventUpdate** with *MANUAL_BROADCASTSEARCH_INPROGRESS* would be provided.
- When Broadcast search (Automatic – started by the TV Module during weak signal) is started, **onDtvEventUpdate** with *AUTO_BROADCASTSEARCH_INPROGRESS* would be provided.
- This would be followed with Intermediatory and concluding events, through ***onDtvBroadcastSearchUpdate***, giving information related to different physical channels searched and reports if any station was found
- If no station was found or Broadcast search Stopped by user, then TV module will retune the formerly tuned last channel, if any was tuned before.
- If a channel is found Broadcast search, search would be concluded and the found channel would be tuned and played.
- Concluding ***onDtvBroadcastSearchUpdate*** would provide information of what kind of station was found (relay or affiliate)
- Both Manual and Automatic Broadcast search can be stopped using ***dtvBroadcastSearch*** with mode as STOP_SEARCH

Pre-Condition:

- TV Module is running, and DTV service is available. There is no Auto preset in progress.

Important:

- During Broadcast Search, other requests (except Broadcast Search) cannot be handled and would result in error response.

Sequence:

For Complete Feature Sequence, refer Broadcast Search

3.2 Set Command

3.2.1 Set Auto Tune Mode

Feature	**TV Settings: Basics**		
Program Epic	CCS2-3565		
API Method Name	***dtvSetAutoTuneMode***		
Parameters	**DtvAutoTuneMode**	***mode***	*AUTO* *FULL_SEGMENT* *ONE_SEGMENT*
	IDtvMethodCallback	***callback***	
Return	**void**		
Result	**IDtvMethodCallback:: onDtvSetAutoTuneModeResult** Refer Dtv Result table for information on result		

Feature Description:

User can configure the component (1seg/12seg) which the Tv Module would tune to.

- Result would be provided through ***onDtvSetAutoTuneModeResult.***
- If *mode* is set as AUTO, Tv Module would tune to 12 seg if reception quality is good, else would switch to 1 seg till the reception improves.
- If *mode* is set as ONE_SEGMENT, Tv Module would tune to 1 seg only, even if 12 seg is available.
- If *mode* is set as FULL_SEGMENT, Tv Module would tune to 12 seg only. If reception is not good, black screen would be shown till the reception improves.
- Set configuration would be persisted.

Pre-Condition:

- TV Module is running, and DTV service is available.

Sequence:

For Complete Feature Sequence, refer Set Auto Tune Mode

3.2.2 Set Auto Broadcast Search Mode

Feature	TV Settings: Basics		
Program Epic	CCS2-3565		
API Method Name	***dtvSetAutoBroadcastSearchMode***		
Parameters	**DtvAutoBroadcastSearchMode**	***mode***	*OFF* *RELAY_STATION_SEARCH* *RELAY_AFFILIATED_STATION_SEARCH*
	IDtvMethodCallback	***callback***	
Return	**void**		
Result	**IDtvMethodCallback:: onDtvSetAutoBroadcastSearchModeResult** Refer Dtv Result table for information on result		

Feature Description:

User can configure the type of channels that can be searched for when the current tuned channel has poor reception / is not receivable anymore.

- Result would be provided through ***onDtvSetAutoBroadcastSearchModeResult.***
- Set configuration would be persisted.

Pre-Condition:

- TV Module is running, and DTV service is available.

Sequence:

For Complete Feature Sequence, refer Set Auto Broadcast Search Mode

3.2.3 Set Auto Event Relay Mode

Feature	**TV Settings: Basics**		
Program Epic	CCS2-3565		
API Method Name	***dtvSetEventRelayMode***		
Parameters	**DtvEventRelayMode**	***mode***	*OFF* *ON*
	IDtvMethodCallback	***callback***	
Return	**void**		
Result	**IDtvMethodCallback:: onDtvSetEventRelayModeResult** Refer Dtv Result table for information on result		

Feature Description:

User can configure the Tv Module, to turn on / turn off automatic tuning of event relay station, in case of an event relay from the broadcaster.

- Result would be provided through ***onDtvSetEventRelayModeResult.***
- Set configuration would be persisted.

Pre-Condition:

- TV Module is running, and DTV service is available.

Sequence:

For Complete Feature Sequence, refer Set Auto Event Relay Mode

3.2.4 Set Closed Caption Mode

Feature	**TV Settings: Basics**		
Program Epic	CCS2-3565		
API Method Name	***dtvSetClosedCaptionMode***		
Parameters	**DtvClosedCaptionMode**	***mode***	*OFF* *CLOSED_CAPTION_1* *CLOSED_CAPTION _2*

	IDtvMethodCallback	***callback***	
Return	**void**		
Result	**IDtvMethodCallback:: onDtvSetClosedCaptionModeResult** Refer Dtv Result table for information on result		
Feature Description:			
User can configure the Tv Module to turn on or off closed caption (subtitle) • Result would be provided through ***onDtvClosedCaptionModeResult.*** • Set configuration would be persisted.			
Pre-Condition:			
• TV Module is running, and DTV service is available.			
Sequence:			
For Complete Feature Sequence, refer Set Closed Caption Mode			

3.2.5 Set Audio Mode

Feature	**TV Settings: Basics**		
Program Epic	CCS2-3565		
API Method Name	***dtvSetAudioMode***		
Parameters	**DtvAudioMode**	***mode***	*MAIN* *SUB* *MAIN_SUB*
	IDtvMethodCallback	***callback***	
Return	**void**		
Result	**IDtvMethodCallback:: onDtvSetAudioModeResult** Refer Dtv Result table for information on result		
Feature Description:			

User can configure the audio mode for Dtv Audio • Result would be provided through ***onDtvSetAudioModeResult.*** • Set configuration would be persisted.
Pre-Condition:
• TV Module is running, and DTV service is available.
Sequence:
For Complete Feature Sequence, refer Set Audio Mode

3.2.6 Set Physical Channel

Feature	**Developer Test Mode (ENG Menu)**		
Program Epic	CCS2-3575		
API Method Name	***dtvSetPhysicalChannel***		
Parameters	**int**	***physicalChannelNumber***	*13 ~ 52*
	IDtvMethodCallback	***callback***	
Return	**void**		
Result	**IDtvMethodCallback:: onDtvSetPhysicalChannelResult** Refer Dtv Result table for information on result		

Feature Description:
User can tune to any physical channel (13 ~ 52) directly • Result would be provided through ***onDtvSetPhysicalChannelResult.*** • Application to ensure that request is sent only for *physicalChannelNumber* 13 to 52 as other request would result in error response. • Current physical channel number would be provided as a part of ***onDtvStatusUpdate*** (Refer Dtv Status Update)
Pre-Condition:
• TV Module is running, and DTV service is available.

Sequence:
For Complete Feature Sequence, refer Set Physical Channel

3.2.7 Set Residential Area

Feature	TV Settings: Prefecture / Zip Code Input		
Program Epic	CCS2-3566		
API Method Name	***dtvSetResidentialArea***		
Parameters	**DtvPresetType**	***presetType***	*HOME* *DRIVE*
	int	***regionId***	*0 (None)* *1~53 (Prefecture)*
	String	***zipcode***	
	IDtvMethodCallback	***callback***	
Return	**void**		
Result	**IDtvMethodCallback:: onDvSetResidentialAreaResult** Refer Dtv Result table for information on result		

Feature Description:

User can set the Region id (prefecture) and zip code for Home Area and Travel Area

- Result would be provided through ***onDvSetResidentialAreaResult***
- It is possible to set either *regionId* only or *zipcode* only or both *regionId* and *zipcode*
- If user selects “None” as the prefecture, *regionId* can be set as 0
- If user selects “Skip” for zipcode, *zipcode* can be set as empty string
- If user selects “None” as prefecture and “Skip” for *zipcode*, ***dtvSetResidentialArea*** should not be sent
- Valid *zipcode* is of 7 digits length
- After receiving ***onDvSetResidentialAreaResult*** as SUCCESS, application should send ***dtvAutoPreset*** with same preset type to scan and store the preset list based on newly set regionId and zipcode information.

Pre-Condition:
• TV Module is running, and DTV service is available.
Sequence:
For Complete Feature Sequence, refer Set Residential Area

3.2.8 Set Audio ES

Feature	**TV Settings: Audio ES**		
Program Epic	**CCS2-3571**		
API Method Name	***dtvSetAudioES***		
Parameters	**int**	***audioESTag***	
	IDtvMethodCallback	***callback***	
Return	**void**		
Result	**IDtvMethodCallback:: onDtvSetAudioESResult** Refer Dtv Result table for information on result		

Feature Description:
User can select the Voice to be played for the current tuned station • Result would be provided through ***onDtvSetAudioESResult*** • When user selects a voice, the *audioESTag* mapped to that voice should be used to send the request • Refer to ***dtvGetAudioES*** for info related to mapping *audioESTag* to voice • Audio ES tags are dynamic and vary between broadcasters • If a channel is re-tuned, changed, lost signal then the selected audio ES would be lost • Audio ES would not be persisted by TV Module
Pre-Condition:
• TV Module is running, and DTV service is available.
Sequence:
For Complete Feature Sequence, refer Audio ES

3.3 Get Command

3.3.1 Get Preset List

Feature	Preset List Retrieve/Update for TV1/TV2		
Program Epic	CCS2-3555		
API Method Name	***dtvGetPresetList***		
Parameters	**DtvPresetType**	***presetType***	*HOME* *DRIVE*
	IDtvMethodCallback	***callback***	
Return	**void**		
Result	**IDtvMethodCallback:: onDtvGetPresetListResult**		
	DtvResult	***result***	Refer Dtv Result table for information on result
	DtvPresetType	***presetType***	
	List<DtvPresetItem>	***presetList***	

Feature Description:

User can fetch the preset list for home or drive preset type.

- Application needs to request ***dtvGetPresetList*** for Home and Drive, once on application start-up.
- Further updates would be provided through ***onDtvPresetListUpdate*** whenever there is any change in the preset list.
- It is possible to fetch preset information for the entire preset list. ***result***, ***presetType*** and ***presetList*** would be provided through ***onDtvGetPresetListResult***.
- If ***result*** is *OK,*
  - ***presetType***: requested preset type
  - ***presetList***: requested preset list information
- **DtvPresetItem** would provide the below information

int	*presetNumber*	
String	*stationName*	

- In case no presets were stored, ***presetList*** with 12 empty **DtvPresetItem** would be provided.
- In case only a few presets were stored, ***presetList*** with 12 **DtvPresetItem** would be provided with the respective index of ***presetList*** updated.
- The data provided through result should be considered only if *result* is OK

Pre-Condition:
• TV Module is running, and DTV service is available. There is no automatic station search or Auto preset in progress.
Sequence:
For Complete Feature Sequence, refer Get Preset List

3.3.2 Get Configure

Feature	**TV Settings: Basics**		
Program Epic	CCS2-3565		
API Method Name	***dtvGetConfigure***		
Parameters	**IDtvMethodCallback**	***callback***	
Return	**void**		
Result	**IDtvMethodCallback:: onDtvGetConfigureResult**		
	DtvResult	***result***	Refer Dtv Result table for information on result
	DtvConfiguration	***config***	
Feature Description:			

- Result would be provided through ***onDtvGetConfigureResult.***
- **DtvConfiguration** would provide the below information

DtvAudioMode	audioMode	*MAIN* *SUB* *MAIN_SUB*
DtvClosedCaptionMode	*captionMode*	*OFF* *CLOSED_CAPTION_1* *CLOSED_CAPTION _2*
DtvAutoTuneMode	*autoTuneMode*	*AUTO* *FULL_SEGMENT* *ONE_SEGMENT*
DtvAutoBroadcastSearch Mode	*searchMode*	*OFF* *RELAY_STATION_SEARCH* *RELAY_AFFILIATED_STATION_SEARCH*
DtvEventRelayMode	*relayMode*	*OFF* *ON*
int	*homeRegionId*	
string	*homeZipcode*	
int	*travelRegionId*	
string	*travelZipcode*	

- The data provided through result should be considered only if *result* is OK

Pre-Condition:

- TV Module is running, and DTV service is available.

Sequence:

For Complete Feature Sequence, refer Get Configure

3.3.3 Get Version Info

Feature	**Developer Test Mode (ENG Menu)**		
Program Epic	CCS2-3575		
API Method Name	***dtvGetVersionInfo***		
Parameters	**IDtvMethodCallback**	***callback***	
Return	**void**		

Result	**IDtvMethodCallback:: onDtv*GetVersion*Result**		
	DtvResult	***result***	Refer Dtv Result table for information on result
	DtvVersionInfo	***info***	

Feature Description:

- Result would be provided through ***onDtvGetVersionResult.***
- **DtvVersionInfo** would provide the below information

String	*hardwareVersion*
String	*bootVersion*
String	*firmwareVersion*
String	*partNumber*
String	*diversityMode*

- The data provided through result should be considered only if *result* is OK

Pre-Condition:

- TV Module is running, and DTV service is available.

Sequence:

For Complete Feature Sequence, refer Get Version Info

3.3.4 Get Program Info

Feature	**Program Information Notification**		
Program Epic	CCS2-3557		
API Method Name	***dtvGetProgramInfo***		
Parameters	**IDtvMethodCallback**	***callback***	
Return	**void**		
Result	**IDtvMethodCallback:: onDtv*GetProgramInfo*Result**		
	DtvResult	***result***	Refer Dtv Result table for information on result

	DtvProgramInfo	*info*	

Feature Description:

- Result would be provided through ***onDtvGetVersionResult.***
- **DtvProgramInfo** would provide the below information

int	presetType	00: Home Preset (TV1) 01: Drive Preset (TV2) FF: None
int	presetNumber	00: Non-Stored Preset 01 ~ 12: Preset Number
int	threeDigitNumber	
int	branchId	
String	stationName	
String	programName	

- The data provided through result should be considered only if *result* is OK

Pre-Condition:

- TV Module is running, and DTV service is available.

Sequence:

For Complete Feature Sequence, refer Get Program Info

3.3.5 Get RMP Manufacturer Id

Feature	TV Settings: RMP Manufacturer ID Display		
Program Epic	CCS2-6560		
API Method Name	***dtvGetRmpManufactuereId***		
Parameters	DtvRMPType	***type***	*HOME,* *DRIVE,* *DEVICE*
	int	***presetNumber***	*00* *01-12*

	IDtvMethodCallback	***callback***	
Return	**void**		
Result	**IDtvMethodCallback:: onDtvGetRmpManufactuereIdResult**		
	DtvResult	***result***	Refer Dtv Result table for information on result
	String	***rmpId***	
Feature Description:			
• Result would be provided through ***onDtvGetRmpManufactuereIdResult*** • RMP Id for Device (RMP Manufacturer Id) and Broadcaster (Generated RMP Manufacturer Id) can be fetched • To Get Device RMP Manufacturer Id, request ***dtvGetRmpManufactuereId*** with **DtvRMPType** as *DEVICE*, ***presetNumber*** as *0* • To Get Broadcaster RMP Manufacturer Id, request ***dtvGetRmpManufactuereId*** with **DtvRMPType** as *HOME/DRIVE* and ***presetNumber*** as the respective preset index • Result **String** *rmpId* would be of the format *X-XXXX-XXXX-XXXX-XXXX-XXXX-XXXX* • The data provided through result should be considered only if result is OK			
Pre-Condition:			
• TV Module is running, and DTV service is available.			
Sequence:			
For Complete Feature Sequence, refer Get RMP Manufacturer ID			

3.3.6 Get Audio ES

Feature	**TV Settings: Audio ES**		
Program Epic	**CCS2-3571**		
API Method Name	***dtvGetAudioES***		
Parameters	**IDtvMethodCallback**	***callback***	
Return	**void**		
Result	**IDtvMethodCallback:: onDtvGetAudioESResult**		

	DtvResult	***result***	Refer Dtv Result table for information on result
	DtvAudioESInfo	***info***	

Feature Description:

- Result would be provided through ***onDtvGetAudioESResult***
- **DtvAudioESInfo** contains the below info

int	*currentAudioESTag*	current selected audio ES tag
int []	*AudioESTags*	list of audio ES tags

- Each audio ES tag should be mapped to a Voice (in the same order as provided)
  e.g., **DtvAudioESInfo** with *currentAudioESTag*:12 and *AudioESTags*: {11,12,13}
  Voice 1: 11
  Voice 2: 12 (current selected)
  Voice 3: 13
- The audio ES tags are dynamic and vary between broadcasters
- The first ES tag is selected and played by default (Voice 1) if no ***setDtvAudioES*** is sent

Pre-Condition:

- TV Module is running, and DTV service is available.

Sequence:

For Complete Feature Sequence, refer Audio ES

3.3.7 Get Dtv Module Status [DEPRECATED]

Feature	**TV Module Status and Error Notifications**		
Program Epic	**CCS2-7661**		
API Method Name	***dtvGetModuleStatus***		
Status	**DEPRECATED :** use ***dtvGetCurrentModuleStatus***		
Parameters	**IDtvMethodCallback**	***callback***	
Return	**void**		
Result	**IDtvMethodCallback:: onDtvGetModuleStatus**		

	DtvModuleStatus	***status***	*LOADING* *READY* *HIGH_TEMPERATURE* *NOT_RESPONSIVE* *INTERNAL_ERROR* *FIRMWARE_UPDATE_INPROGRESS* *FIRMWARE_UPDATE_FAILURE*

Feature Description:

- Result would be provided through **onDtvGetModuleStatus**
- ***dtvGetModuleStatus*** can be requested by Application once on start-up and follow further updates based on ***onDtvModuleStatusUpdate***

LOADING	TV Module startup in progress
READY	TV Module up and ready to process all requests
HIGH_TEMPERATURE	TV Module has high temperature
NOT_RESPONSIVE	TV Module not responding
INTERNAL_ERROR	TV Module has internal error
FIRMWARE_UPDATE_INPROGRESS	TV Module Firmware Update in progress
FIRMWARE_UPDATE_FAILURE	TV Module Firmware Update failed

- Dtv functionality would be available and can be requested only when the *DtvModuleStatus* is *READY*
- During other states, TV Menu and TV Settings button should be greyed out and Preset List should not be visible for user to access
- Appropriate Advisories should be shown when *DtvModuleStatus* is not *READY* and cleared on receiving *READY*

Pre-Condition:

- TV Module is running, and DTV service is available.

Sequence:

For Complete Feature Sequence, refer Module Status

3.3.8 Get Dtv Antenna State

Feature	Developer Test Mode (ENG Menu)		
Program Epic	CCS2-3575		
API Method Name	***dtvGetAntennaState***		
Parameters	**DtvAntenna**	***antenna***	DTV_ANTENNA_1 DTV_ANTENNA_2 DTV_ANTENNA_3 DTV_ANTENNA_4 DTV_ANTENNA_ALL
Return	**List<DtvAntennaStatus>**		

Feature Description:

- **DtvAntennaStatus** would provide the below information

DtvAntenna	antenna	DTV_ANTENNA_1 DTV_ANTENNA_2 DTV_ANTENNA_3 DTV_ANTENNA_4
DtvAntennaState	antennaState	DTV_ANTENNA_STATE_INVALID DTV_ANTENNA_STATE_OK DTV_ANTENNA_STATE_OPEN DTV_ANTENNA_STATE_SHORT DTV_ANTENNA_DEACTIVATED DTV_ANTENNA_DEACTIVATED_LOW_VLTG DTV_ANTENNA_DEACTIVATED_HIGH_VLTG

- **dtvGetAntennaState** with DTV_ANTENNA_ALL would provide List<**DtvAntennaStatus>** with state for all available Dtv Antenna

Pre-Condition:

- TV Module is running, and DTV service is available.

Sequence:

For Complete Feature Sequence, refer Dtv Antenna State

3.3.9 Get Current Dtv Module Status

Feature	TV Module Status and Error Notifications

Program Epic	CCS2-7661		
API Method Name	***dtvGetCurrentModuleStatus***		
Parameters	**void**	***void***	
Return	**DtvModuleStatus**	***status***	*LOADING* *READY* *HIGH_TEMPERATURE* *NOT_RESPONSIVE* *INTERNAL_ERROR* *FIRMWARE_UPDATE_INPROGRESS* *FIRMWARE_UPDATE_FAILURE*

Feature Description:

- ***dtvGetCurrentModuleStatus*** can be requested by Application for
  - ***onServiceConnected***

  and follow further updates based on ***onDtvModuleStatusUpdate***

LOADING	TV Module startup in progress
READY	TV Module up and ready to process all requests
HIGH_TEMPERATURE	TV Module has high temperature
NOT_RESPONSIVE	TV Module not responding
INTERNAL_ERROR	TV Module has internal error
FIRMWARE_UPDATE_INPROGRESS	TV Module Firmware Update in progress
FIRMWARE_UPDATE_FAILURE	TV Module Firmware Update failed

- Dtv functionality would be available and can be requested only when the *DtvModuleStatus* is *READY*
- During other states, TV Menu and TV Settings button should be greyed out and Preset List should not be visible for user to access
- Appropriate Advisories should be shown when *DtvModuleStatus* is not *READY* and cleared on receiving *READY*

Pre-Condition:

- TV Module is running, and DTV service is available.

Sequence:

For Complete Feature Sequence, refer Module Status

3.4 DTV Events

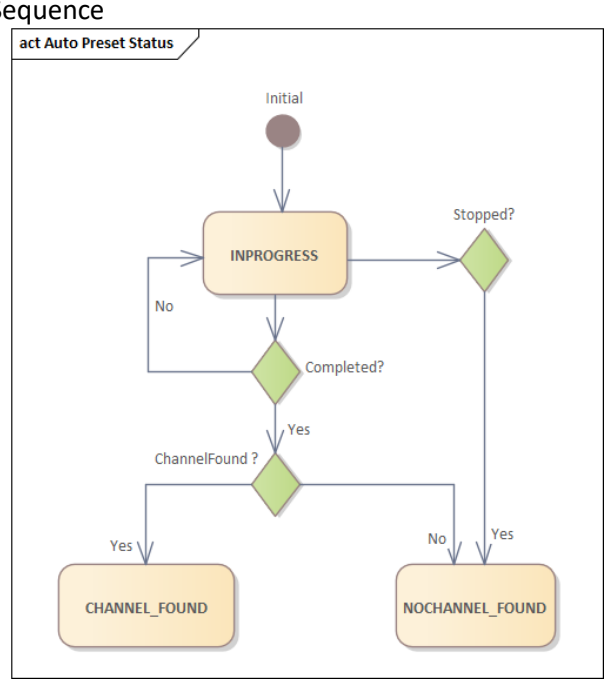
3.4.1 Auto Preset Update

Feature	**Auto Preset for TV1/TV2**		
Program Epic	CCS2-3554		
API Event Name	***onDtvAutoPresetUpdate***		
Parameters	**DtvAutoPresetStatus**	***status***	*INPROGRESS* *CHANNEL_FOUND* *NOCHANNEL_FOUND*
	boolean	***isDetected***	
	int	***channelNumber***	*00* *13~52 (physical channel number)*
Return	**void**		
Events to Follow	**(optional) IDtvEventsCallback:: onDtvPresetListUpdate**		

Feature Description:

User can perform Auto Preset to scan entire band and store receivable channels into TV1 or TV2 preset list. Up to 12 stations can be stored into each preset list.

- When Auto Preset is started successfully, Intermediatory and concluding events would be provided, through ***onDtvAutoPresetUpdate***, giving information related to each physical channels from 13 to 52 scanned and reports if any station was found or not.
- When **DtvAutoPresetStatus** is INPROGRESS,
  - ***isDetected*** == True**:** valid signal was detected on the physical channel number
  - ***isDetected*** == False**:** no valid signal was detected on the physical channel number
  - ***channelNumber***: current scanned physical channel number (13~52)
- When **DtvAutoPresetStatus** is not INPROGRESS, ignore ***isDetected***, ***channelNumber***
- When ***dtvAutoPreset*** *STOP* is requested, ***onDtvAutoPresetUpdate*** with **DtvAutoPresetStatus** as *NOCHANNEL_FOUND* would be provided.

Pre-Condition:
• TV Module is running, and service is available • Application has registered to receive DTV Events by registering through ***dtvRegisterCallback*** • ***dtvAutoPreset*** called and ***onAutoPreset*** Result was OK
Sequence:
DtvAutoPresetStatus Transition Sequence    For Complete Feature Sequence, refer Auto Preset

3.4.2 Channel Seek Update

Feature	**Channel Seek Up/Down**		
Program Epic	CCS2-1617		
API Event Name	***onDtvChannelSeekUpdate***		
Parameters	**DtvSeekStatus**	***status***	*INPROGRESS* *CHANNEL_FOUND* *NOCHANNEL_FOUND*

	boolean	***isDetected***	
	int	***channelNumber***	*00* *13~52 (physical channel number)*
Return	**void**		

Feature Description:

User can perform Channel Seek to search the band, to find other receivable channels.

- When Channel Seek is started successfully, Intermediatory and concluding events would be provided, through ***onDtvChannelSeekUpdate***, giving information related to different physical channels searched and reports if any channel was found
- When **DtvSeekStatus** is INPROGRESS,
  - ***isDetected*** == True**:** valid signal was detected for the physical channel number
  - ***isDetected*** == False**:** no valid signal was detected for the physical channel number
  - ***channelNumber***: current scanned physical channel number (13~52)
- When **DtvSeekStatus** is not INPROGRESS, ignore ***isDetected***, ***channelNumber***

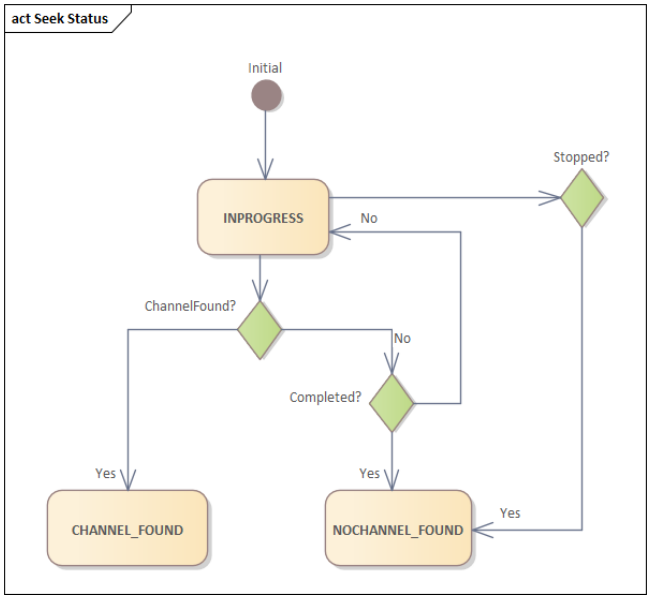
When ***dtvSeekChannel*** STOP is requested, ***onDtvChannelSeekUpdate*** with **DtvSeekStatus** as *NOCHANNEL_FOUND* would be provided.

Pre-Condition:

- TV Module is running, and service is available
- Application has registered to receive DTV Events by registering through ***dtvRegisterCallback***
- ***dtvSeekChannel*** called and ***onDtvSeekChannelResult*** was OK

Sequence:

DtvSeekStatus Transition Sequence



For Complete Feature Sequence, refer Channel Seek

3.4.3 Program Infomation Update

Feature	**Program Information Notification**		
Program Epic	CCS2-3557		
API Event Name	***onDtvProgramInfoUpdate***		
Parameters	**DtvProgramInfo**	***info***	
Return	**void**		
Feature Description:			

- ***onDtvProgramInfoUpdate*** would be provided whenever there is an update in the Program Information, related to the current tuned channel
- **DtvProgramInfo** would provide the below information

int	presetType	00: Home Preset (TV1) 01: Drive Preset (TV2) FF: None
int	presetNumber	00: Non-Stored Preset 01 ~ 12: Preset Number
int	threeDigitNumber	
int	branchId	
String	stationName	
String	programName	

- If current tuned channel is not stored, presetType would be FF and presetNumber would be 00
- After initialization, ***onDtvProgramInfoUpdate*** would be provided with all fields cleared

Pre-Condition:

- TV Module is running, and service is available. Any channel is tuned by TV module.
- Application has registered to receive DTV Events by registering through ***dtvRegisterCallback***

3.4.4 Preset List Update

Feature	**Preset List Retrieve/Update for TV1/TV2**		
Program Epic	CCS2-3555		
API Event Name	***onDtvPresetListUpdate***		
Parameters	**DtvPresetType**	***presetType***	
	List<DtvPresetItem>	***presetList***	
Return	**void**		

Feature Description:

- ***onDtvPresetListUpdate*** would be provided in the below use cases
  - startup (both TV1 and TV2 preset list)
  - after **AutoPreset** if new channel was found (preset list for requested preset type during Auto Preset)
  - after **storePresetChannel** was successful
- **DtvPresetType** would provide info if the preset list is for Home or Drive preset type.
- The entire preset list would be provided in the update.
- If no presets were stored, ***presetList*** with 12 empty **DtvPresetItem** would be provided.
- In case only a few presets were stored, ***presetList*** with 12 **DtvPresetItem** would be provided with the respective index of ***presetList*** updated.
- **DtvPresetItem** would provide the below information

int	presetNumber	
String	stationName	

Pre-Condition:

- TV Module is running, and service is available. Any channel is tuned by TV module.
- Application has registered to receive DTV Events by registering through ***dtvRegisterCallback***

3.4.5 Signal Status Update [DEPRICATED]

Feature	Signal Status (Weak / No signal) / No Service Notification		
Program Epic	CCS2-3573		
Status	***DEPRICATED***: use ***onDtvEventUpdate*** instead		
API Event Name	***onDtvSignalStatusUpdate***		
Parameters	**DtvSignalStatus**	***status***	*OK* *WEAKSIGNAL* *NOSIGNAL* *NOBROADCAST* *TRMPERROR* *RMPERROR_EC21* *RMPERROR_EC22* *RMPERROR_EC23*

Return	void	

Feature Description:

- ***onDtvSignalStatusUpdate***
- would be provided whenever there is a change in the reception of currently played channel.
- Information related to **DtvSignalStatus** is as below

OK	Default state, no error
WEAKSIGNAL	Channel tuned but signal lost
NOSIGNAL	Channel tuning error, signal not found
NOBROADCAST	Channel not being broadcasted
TRMPERROR	TRMP information is invalid
RMPERROR_EC21	
RMPERROR_EC22	
RMPERROR_EC23	

Pre-Condition:

- TV Module is running, and service is available. Any channel is tuned by TV module.
- Application has registered to receive DTV Events by registering through ***dtvRegisterCallback***

3.4.6 Dtv Status Update

Feature	Antenna Bars Level (0/1/2/3) Notification		
Program Epic	CCS2-3558		
API Event Name	***onDtvStatusUpdate***		
Parameters	DtvStatus	***status***	
Return	void		
Feature Description:			

Status of few key parameters would be provided through ***onDtvStatusUpdate***

- **DtvStatus** would provide the below information.

DtvTuningStatus	tuningStatus	Tuning Status • IDLE: Idle • TUNING: Tuning
DtvServiceType	serviceType	Service Type (Now Playing Service Type) • ISDB_T_FULLSEG: ISDB-T Fullseg • ISDB_T_ONESEG: ISDB-T Oneseg • NONE: None (Not Tuning)
DtvAntennaLevel	antennaLevel	Antenna Level • DTV_ANTENNA_LEVEL0: Antenna Level 0 • DTV_ANTENNA_LEVEL1: Antenna Level 1 • DTV_ANTENNA_LEVEL2: Antenna Level 2 • DTV_ANTENNA_LEVEL3: Antenna Level 3
int	physicalChannelNumber	00, 13 ~ 52
DtvEpgBmlStatus	epgbmlStatus	EPG/BML Status • NO_EPG_BML: No EPG/BML displayed • UPDATING: Updating • EPG_FULLSEG: EPG List (full seg) • EPG_ONESEG: EPG List (one seg) • EPG_INFO_FULLSEG: EPG Information (full seg) • EPG_INFO_ONESEG: EPG Information (one seg) • BML_DISPLAY: BML Display

- **tuningStatus** provides information related to the current state of TV Tuner.
  - *IDLE:* not tuned to any channel
  - *TUNING:* tuned to a channel
- **serviceType** shows current Service Type (Now Playing Service Type) – One Seg or Full Seg or Not Tuning.
- **antennaLevel** provides information related to the current strength of signal received through the antenna. “Antenna Level” indicates the number of antennas on the display.
  - Antenna Level 0 is bad signal.
  - Antenna Level 3 is good signal.
  - If “Tuning Status (tuningStatus)” is idle, Antenna Level is 0.
- **physicalChannelNumber** provides the current physical channel number tuned.
- **epgbmlStatus** provides information related to the status of EPG/BML Screen.

For Complete Feature Sequence, refer Dtv Status Update

Pre-Condition:
• TV Module is running, and service is available. • Application has registered to receive DTV Events by registering through ***dtvRegisterCallback***

3.4.7 Reception Monitor Update (Test Mode)

Feature	Developer Test Mode (ENG Menu)		
Program Epic	CCS2-3575		
API Event Name	***OnDtvReceptionMonitorUpdate***		
Parameters	**ReceptionInfo**	***info***	
Return	**void**		
Feature Description:			

Status of few key reception parameters would be provided through ***OnDtvReceptionMonitorUpdate***

- **ReceptionInfo** would provide the below information

String	*ant1_CN*	Signal / Noise of Antenna 1 in dB
String	*ant2_CN*	Signal / Noise of Antenna 2 in dB
String	*ant3_CN*	Signal / Noise of Antenna 3 in dB
String	*ant4_CN*	Signal / Noise of Antenna 4 in dB
String	*combined_CN*	Signal / Noise of Combined Tuner in dB
LayerInfo	*layerA*	
LayerInfo	*layerB*	
LayerInfo	*layerC*	

- **LayerInfo** would provide the below information

boolean	*isActive*	is layer Active or Not
String	*BER*	Bit Error Rate in %
String	*PER*	Packet Error Rate in %

- If a layer is not active, *isActive* would be set to false and BER, PER would be empty string.
- When ***dtvReceptionMonitorSession*** with *START* state is requested, ***OnDtvReceptionMonitorUpdate*** would be provided every second.
- When ***dtvReceptionMonitorSession*** with *STOP* state is requested, ***OnDtvReceptionMonitorUpdate*** updates would be stopped.

Formula for Calculation REC:

Reception rate (%) = {1- (number of data items with PER 1% or more) / (time elapsed)} x 100

Example:

User press Start in Reception Monitor Screen

1st Monitor Status update

Duration: 1s	PER	Calculated REC
Layer A	0.6	{1 – 0 / 1} * 100 = 100 %
Layer B	1.1	{1 – 1 / 1} * 100 = 0 %
Layer C	0	{1 – 0 / 1} * 100 = 100 %

2nd Monitor Status update

Duration: 2s	PER	Calculated REC
Layer A	1.2	{1 – 1 / 2} * 100 = 50 %
Layer B	1.3	{1 – 2 / 2} * 100 = 0 %
Layer C	0	{1 – 0 / 2} * 100 = 100 %

3rd Monitor Status update

Duration: 3s	PER	Calculated REC	Observation
Layer A	1.8	{1 – 2 / 3} * 100 = 67.33 %	Intermediate reception, some packets missed
Layer B	1.5	{1 – 3 / 3} * 100 = 0 %	Very Poor Reception, all packets missed
Layer C	0	{1 – 0 / 3} * 100 = 100 %	Very Good Reception, no or very less packets missed

For Complete Feature Sequence, refer Reception Monitor (Test Mode)

Pre-Condition:
• TV Module is running, and service is available • Application has registered to receive DTV Events by registering through ***dtvRegisterCallback***

3.4.8 Dtv Firmware Update

Feature	**Developer Test Mode (ENG Menu)**		
Program Epic	CCS2-3575		
API Event Name	***onDtvFirmwareUpdate***		
Parameters	**DtvFirmwareUpdateStatus**	***status***	*START* *FILETRANSFER_INPROGRESS* *FLASH_INPROGRESS* *FLASH_SUCCESS* *COMPLETE* *ABORTED* *FAILURE_INVALID_FILENAME* *FAILURE_INVALID_DATA* *FAILURE_INVALID_FILESIZE* *FAILURE_TIMEOUT*
Return	**void**		

Feature Description:
Progress of firmware update would be provided through this interface**.** • Once Firmware Update is complete (SWUPDATE_COMPLETE), Application must **Get Program Info** and **Get Preset List** (Home Area, Travel Area) again. For Complete Feature Sequence, refer Dtv Firmware Update
Pre-Condition:
• TV Module is running, and service is available • Application has registered to receive DTV Events by registering through ***dtvRegisterCallback***

3.4.9 Dtv EPG Update

Feature	EPG (8 Days Program Schedule for multiple channels)		
Program Epic	CCS2-3552		
API Event Name	***onDtvEpgUpdate***		
Parameters	DtvEpgUpdateStatus	***status***	*INPROGRESS* *SUCCESS* *CANCELLED*
Return	void		

Feature Description:

Status of EPG Update started by user would be provided through ***onDtvEpgUpdate***

- When EPG Update is *INPROGRESS* and user wants to exit EPG, first ***dtvDisplayEpg*** with *CANCEL_UPDATE* should be sent, followed with ***dtvDisplayEpg*** with *BACK* or *CLOSE_ALL*
- After receiving ***onDtvEpgUpdate*** as *SUCCESS*, application should request **dtvDisplayEpg** with ***DtvEpgOperation*** as *DISPLAYEPG_LIST* to update the EPG button status

Pre-Condition:

- TV Module is running, and service is available
- Application has registered to receive DTV Events by registering through ***dtvRegisterCallback***
- User has selected Update in EPG Program Guide

Sequence:

For Complete Feature Sequence, refer EPG

3.4.10 Broadcast Search Update

Feature	Manual Foreground Broadcast Search (Relay/Affiliated Station) / Automatic Foreground Broadcast Search (Relay/Affiliated Station)
Program Epic	CCS2-3560 / CCS2-3561
API Event Name	***onDtvBroadcastSearchUpdate***

Parameters	**DtvBroadcastSearchStatus**	***status***	*INPROGRESS* *COMPLETE_NOSTATION_FOUND* *COMPLETE_RELAYSTATION_FOUND* *COMPLETE_AFFILIATEDSTATION_FOUND*
	boolean	***isDetected***	
	int	***channelNumber***	*00* *13~52 (physical channel number)*
Return	**void**		

Feature Description:

- When Broadcast Search (Manual / Automatic) is ongoing Intermediatory and concluding events would be provided, through **DtvBroadcastSearchStatus**
- When **DtvBroadcastSearchStatus** is INPROGRESS,
  - ***isDetected*** == True**:** valid signal was detected for the physical channel number
  - ***isDetected*** == False**:** no valid signal was detected for the physical channel number
  - ***channelNumber***: current scanned physical channel number (13~52)
- When **DtvBroadcastSearchStatus** is not INPROGRESS, ignore ***isDetected***, ***channelNumber***
- If **DtvBroadcastSearchStatus**
  - COMPLETE_NOSTATION_FOUND: No channel was found during search. TV Module would tune to back previous playing channel if any
  - COMPLETE_RELAYSTATION_FOUND: Relay station was found during search and TV Module would tune to the relay station.
  - *COMPLETE_AFFILIATEDSTATION_FOUND:* Affiliate station was found during search and TV Module would tune to the affiliated station.
- When **dtvBroadcastSearch** STOP_SEARCH is requested, ***onDtvBroadcastSearchUpdate*** with **DtvBroadcastSearchStatus** as *COMPLETE_NOSTATION_FOUND* would be provided as the conclusive event

Pre-Condition:

- TV Module is running, and service is available
- Application has registered to receive DTV Events by registering through ***dtvRegisterCallback***

Sequence:

For Complete Feature Sequence, refer Broadcast Search

3.4.11 EWS Update

Feature	EWS (Emergency Warning Service) Notification		
Program Epic	CCS2-3563		
API Event Name	***onDtvEwsEventUpdate***		
Parameters	DtvEwsEventLevel	*level*	*END* *LEVEL_1* *LEVEL_2* *TEST*
	int	***threeDigitNumber***	*000~199 (Full seg)* *600~799 (One seg)*
Return	void		

Feature Description:

Emergency Warning Updates would be provided through this interface whenever such updates are received from the broadcaster.

- Actions to be performed by application on receiving **onDtvEwsEventUpdate**

DtvEwsEventLevel	
LEVEL_1	Application should check if provided threeDigitNumber is different from the current playing threeDigitNumber (provided through ***onDtvProgramInfoUpdate***). If yes, Application should tune to the provided *threeDigitNumber* using ***dtvDirectChannelSelect***
LEVEL_2 *TEST*	Application should show Emergency Warning popup to user. If user selects "Yes", application should check if provided threeDigitNumber is different from the current playing threeDigitNumber (provided through ***onDtvProgramInfoUpdate***). If yes, Application should tune to the provided *threeDigitNumber* using ***dtvDirectChannelSelect***
END	If Emergency Warning popup shown, application should hide the popup

Pre-Condition:

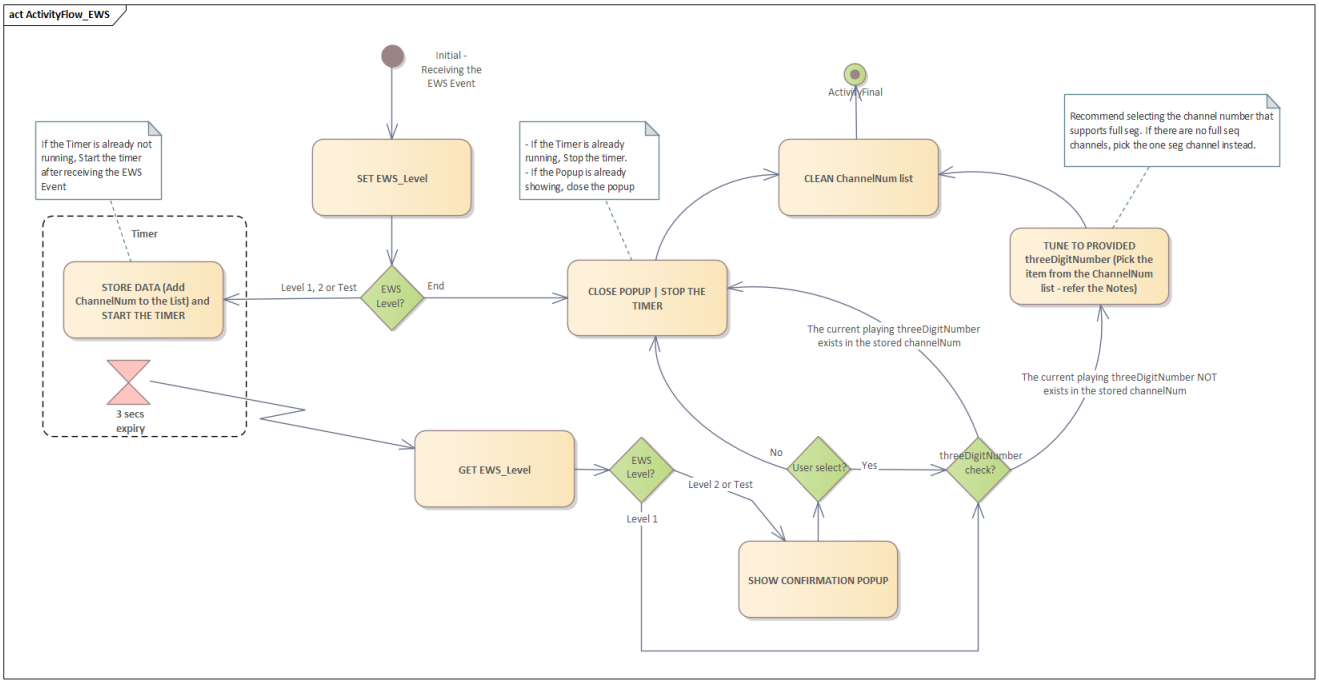
- TV Module is running, and service is available.
- Application has registered to receive DTV Events by registering through ***dtvRegisterCallback***

Important:

- When displaying EPG/Program Information or BML, it is necessary to close EPG/Program Information or BML before switching to EWS service.

Sequence:

The Application should follow the activity diagram below to make sure the EWS feature works as specs:



For Complete Feature Sequence, refer EWS

3.4.12 Dtv Event Update

Feature	**Signal Status (Weak / No signal) / No Service Notification / Manual Foreground Broadcast Search (Relay/Affiliated Station) / Automatic Foreground Broadcast Search (Relay/Affiliated Station)**		
Program Epic	CCS2-3573 / CCS2-3560 / CCS2-3561		
API Event Name	***onDtvEventUpdate***		
Parameters	**DtvEvent**	***event***	*NO_ERROR* *WEAKSIGNAL* *NOSIGNAL* *NOBROADCAST* *TRMP_INVALID* *RMPERROR_EC21* *RMPERROR_EC22* *RMPERROR_EC23* *TUNER_ERROR* *TUNE_INPROGRESS* *TUNE_LASTCHANNEL_INPROGRESS* *SEEK_INPROGRESS* *MANUAL_BROADCASTSEARCH_INPROGRESS* *AUTO_BROADCASTSEARCH_INPROGRESS* *TUNE_COMPLETE*
Return	**void**		
Feature Description:			

- ***onDtvEventUpdate*** would be provided whenever there is a change in the reception of currently played channel

NO_ERROR	Default state, no error	No Advisory
WEAKSIGNAL	Channel tuned but signal lost	Weak Signal Advisory
NOSIGNAL	Channel tuning error, signal not found	No Signal Advisory
NOBROADCAST	Channel not being broadcasted	No Broadcast Advisory
TRMP_INVALID	TRMP information is invalid	No Advisory
RMPERROR_EC21		The reception control data is not set. Please wait for a moment. (It may take a maximum of 30 seconds.)
RMPERROR_EC22		This channel cannot be viewed. Please contact the customer center of the channel
RMPERROR_EC23		This program cannot be viewed with this receiver. Please contact the customer center of the channel
TUNER_ERROR		No Advisory
TUNE_INPROGRESS	Tuning to new channel	No Advisory
TUNE_LASTCHANNEL_INPROGRESS	Tuning to last channel	No Advisory
SEEK_INPROGRESS	Channel Seek Started by user	Show Seek Popup
MANUAL_BROADCASTSEARCH_INPROGRESS	Broadcast Search started by user	Show Affiliated Search Popup
AUTO_BROADCASTSEARCH_INPROGRESS	Broadcast Search started by TV Module	Show Auto Station Search Popup

- Application can show “Seek Popup” popup when **onDtvEventUpdate** with *SEEK_INPROGRESS* is received
- Application can show “Affiliated Station Search” popup when **onDtvEventUpdate** with *MANUAL_BROADCASTSEARCH_INPROGRESS* is received
- Application can show “Auto Station Search” popup when **onDtvEventUpdate** with *AUTO_BROADCASTSEARCH_INPROGRESS* is received
- In case of *TUNER_ERROR / TRMP_INVALID,* current advisory can be cleared. Appropriate error notification would be given through **onDtvModuleStatusUpdate**

Pre-Condition:

- TV Module is running, and service is available.
- Application has registered to receive DTV Events by registering through ***dtvRegisterCallback***

3.4.13 Dtv Module Status Update

Feature	**TV Module Status and Error Notifications**		
Program Epic	**CCS2-7661**		
API Event Name	***onDtvModuleStatusUpdate***		
Parameters	**DtvModuleStatus**	***status***	*LOADING* *READY* *HIGH_TEMPERATURE* *NOT_RESPONSIVE* *INTERNAL_ERROR* *FIRMWARE_UPDATE_INPROGRESS* *FIRMWARE_UPDATE_FAILURE*
Return	**void**		

Feature Description:

- ***onDtvModuleStatusUpdate*** would be provided whenever there is a change in the state of TV Module

LOADING	TV Module startup in progress
READY	TV Module up and ready to process all requests
HIGH_TEMPERATURE	TV Module has high temperature
NOT_RESPONSIVE	TV Module not responding
INTERNAL_ERROR	TV Module has internal error
FIRMWARE_UPDATE_INPROGRESS	TV Module Firmware Update in progress
FIRMWARE_UPDATE_FAILURE	TV Module Firmware Update failed

- Dtv functionality would be available and can be requested only when the *DtvModuleStatus* is *READY*
- During other states, TV Menu and TV Settings button should be greyed out and Preset List should not be visible for user to access
- Appropriate Advisories should be shown when *DtvModuleStatus* is not *READY* and cleared on receiving *READY*

Pre-Condition:

- Application has registered to receive DTV Events by registering through ***dtvRegisterCallback***

3.4.14 Dtv Antenna State Update

Feature	**Developer Test Mode (ENG Menu)**		
Program Epic	CCS2-3575		
API Event Name	***onDtvAntennaStateUpdate***		
Parameters	**List<DtvAntennaStatus>**	***antennaStatus***	
Return	**void**		

Feature Description:

- ***onDtvAntennaStateUpdate*** would be provided whenever there is a change in the state of any Dtv Antenna
- **DtvAntennaStatus** would provide the below information

DtvAntenna	antenna	DTV_ANTENNA_1 DTV_ANTENNA_2 DTV_ANTENNA_3 DTV_ANTENNA_4
DtvAntennaState	antennaState	DTV_ANTENNA_STATE_INVALID DTV_ANTENNA_STATE_OK DTV_ANTENNA_STATE_OPEN DTV_ANTENNA_STATE_SHORT DTV_ANTENNA_DEACTIVATED DTV_ANTENNA_DEACTIVATED_LOW_VLTG DTV_ANTENNA_DEACTIVATED_HIGH_VLTG

Pre-Condition:

- Application has registered to receive DTV Events by registering through ***dtvRegisterCallback***

3.5 DTV Service Connection Noitification

3.5.1 Service Connected

API Event Name	***onServiceConnected***

Parameters	void	*void*	
Return	void		

Feature Description:

- ***onServiceConnected*** would be notified when com.bosch.dtv.service is available and ready to provide service
- This notification would be provided
  - On start-up after DtvManager instance is created and com.bosch.dtv.service is available
  - Upon recovery after previous ***onServiceDisconnected*** notification
- On receiving this notification application should
  - Register for dtv events through ***dtvRegisterCallback***
  - Initiate the start-up sequence handling (Refer Dtv Service Connected – Handling)

For sequence, refer Dtv Service Connection Notification

Pre-Condition:

- Application has passed a valid ***IDtvServiceConnectionListener*** instance while creating the DtvManager instance

3.5.2 Service Disconnected

API Event Name	*onServiceDisconnected*		
Parameters	void	*void*	
Return	void		

Feature Description:

- ***onServiceDisconnected*** would be notified when com.bosch.dtv.service is unavailable to provide service
- This notification would be provided
  - When onBinderDied notification is received for com.bosch.dtv.service binder
- On receiving this notification application should
  - Initiate the disconnected sequence handling (Refer Dtv Service Disconnected – Handling)

For sequence, refer Dtv Service Connection Notification

Pre-Condition:
• Application has passed a valid ***IDtvServiceConnectionListener*** instance while creating the DtvManager instance

3.6 DTV Service Status Notification

3.6.1 Service Suspended

Feature			
Program Epic			
API Event Name	***onSuspendEnter***		
Parameters	**void**	***void***	
Return	**void**		
Feature Description:			
• ***onSuspendEnter*** would be notified when com.bosch.dtv.service receives SUSPEND_ENTER state from CarPowerManager • On receiving this notification application should - Initiate the suspendEnter sequence handling (Refer Suspend)			
Pre-Condition:			
• Application has registered to receive Dtv Service Notification by registering through ***dtvRegisterServiceStatusCallback***			

3.6.2 Service Resume

Feature			
Program Epic			
API Event Name	***onSuspendExit***		
Parameters	**void**	***void***	

Return	void	
Feature Description:		
• ***onSuspendExit*** would be notified when com.bosch.dtv.service receives SUSPEND_EXIT state from CarPowerManager • On receiving this notification application should - Initiate the suspendExit sequence handling (Refer Resume)		
Pre-Condition:		
• Application has registered to receive Dtv Service Notification by registering through ***dtvRegisterServiceStatusCallback***		

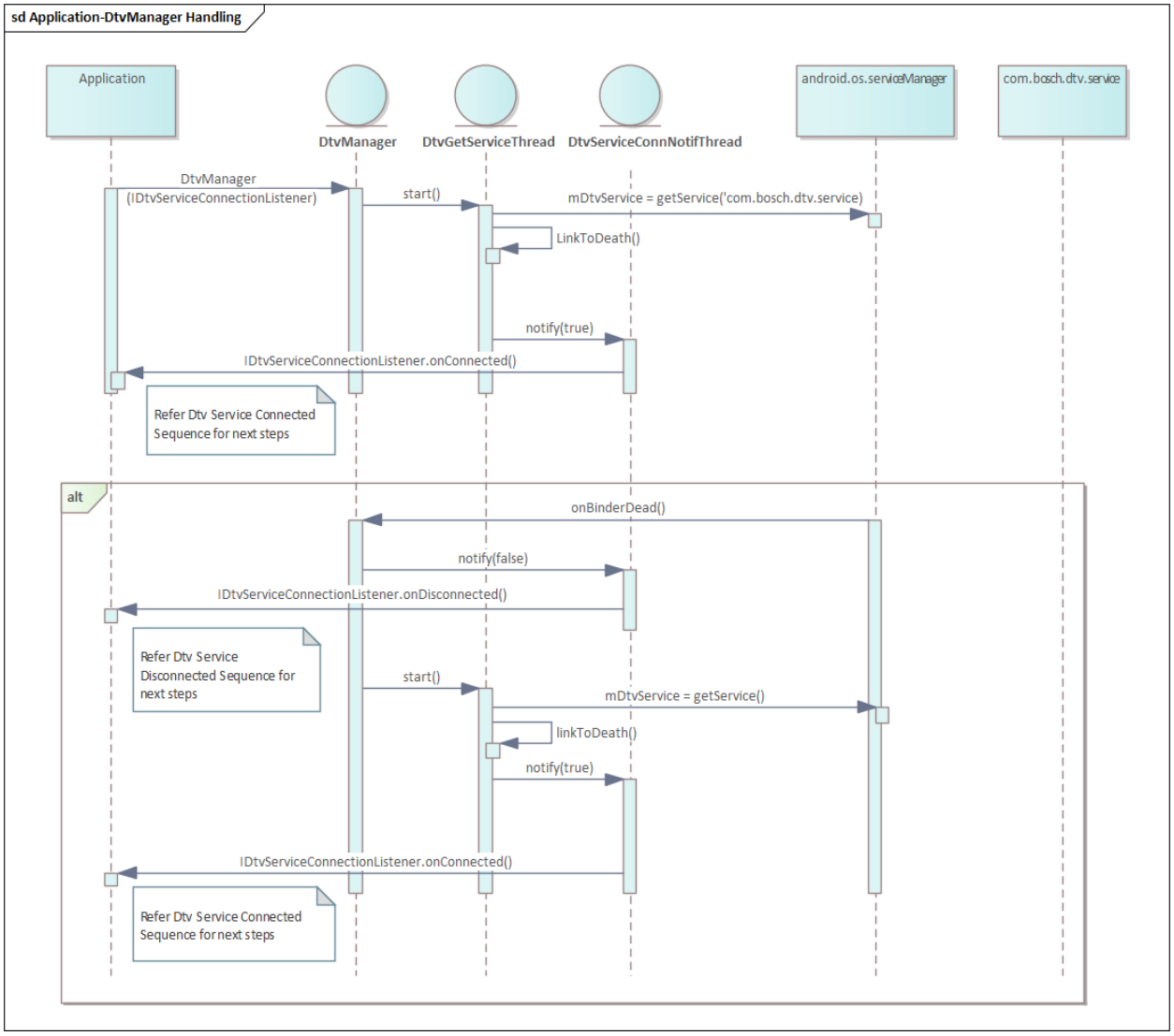
3.7 DTV Results

OK	Success
INVALID_PARAM	
INVALID_DATA	
INVALID_LENGTH	
NO_SERVICE	
NOT_SUPPORT	
WEAK_SIGNAL	
CHANNEL_NOT_RECEIVABLE	
CHANNEL_NOT_BROADCASTED	
INVALID_TRMP	
INVALID_STATUS	
RMP_EC21	
RMP_EC22	
RMP_EC23	
UNKNOWN_ERROR	

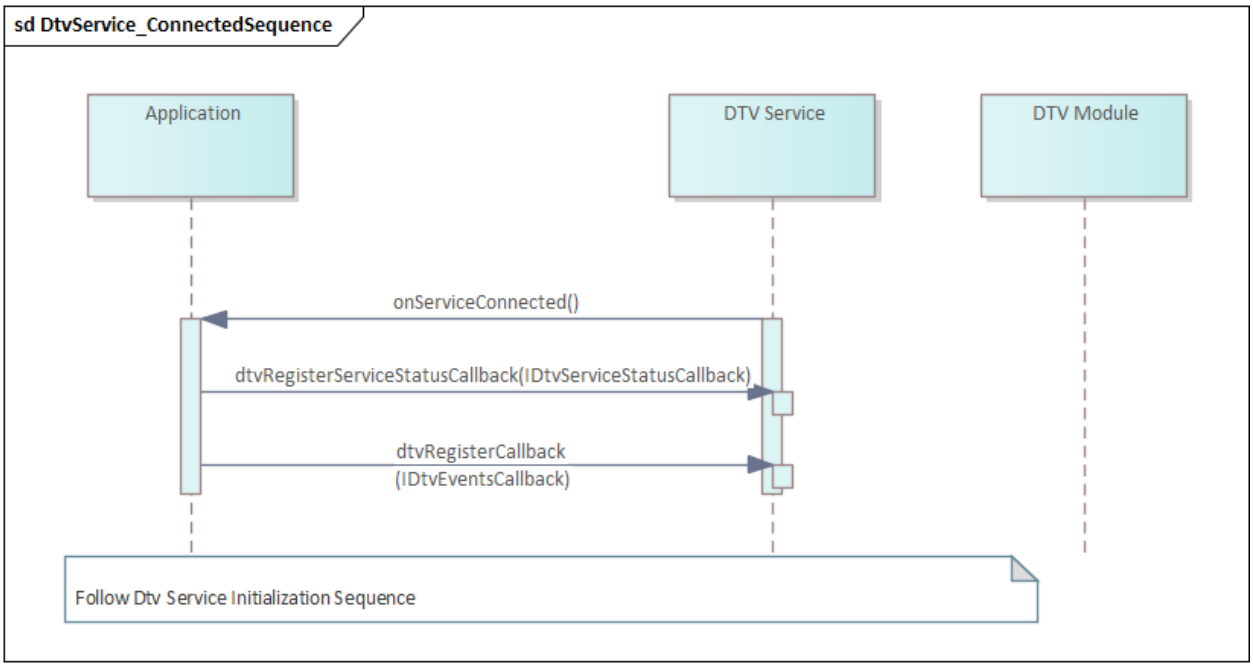
4 Sequences

4.1 Startup Communication Sequence

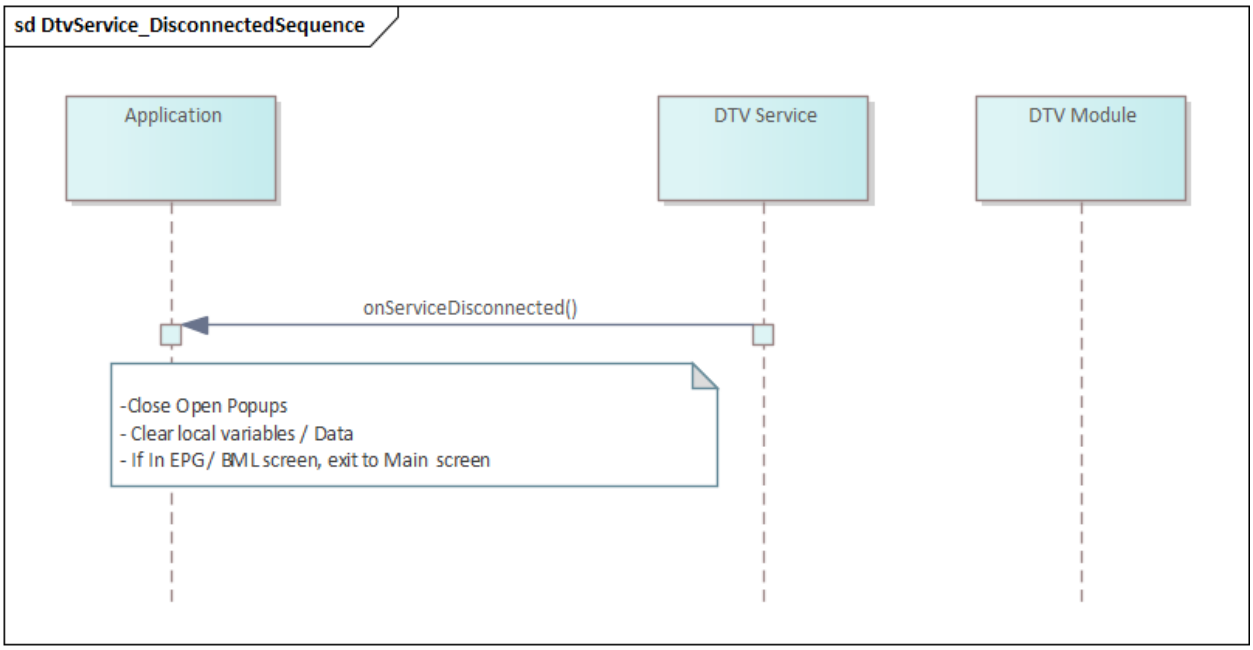
4.1.1 Dtv Service Connection Notification



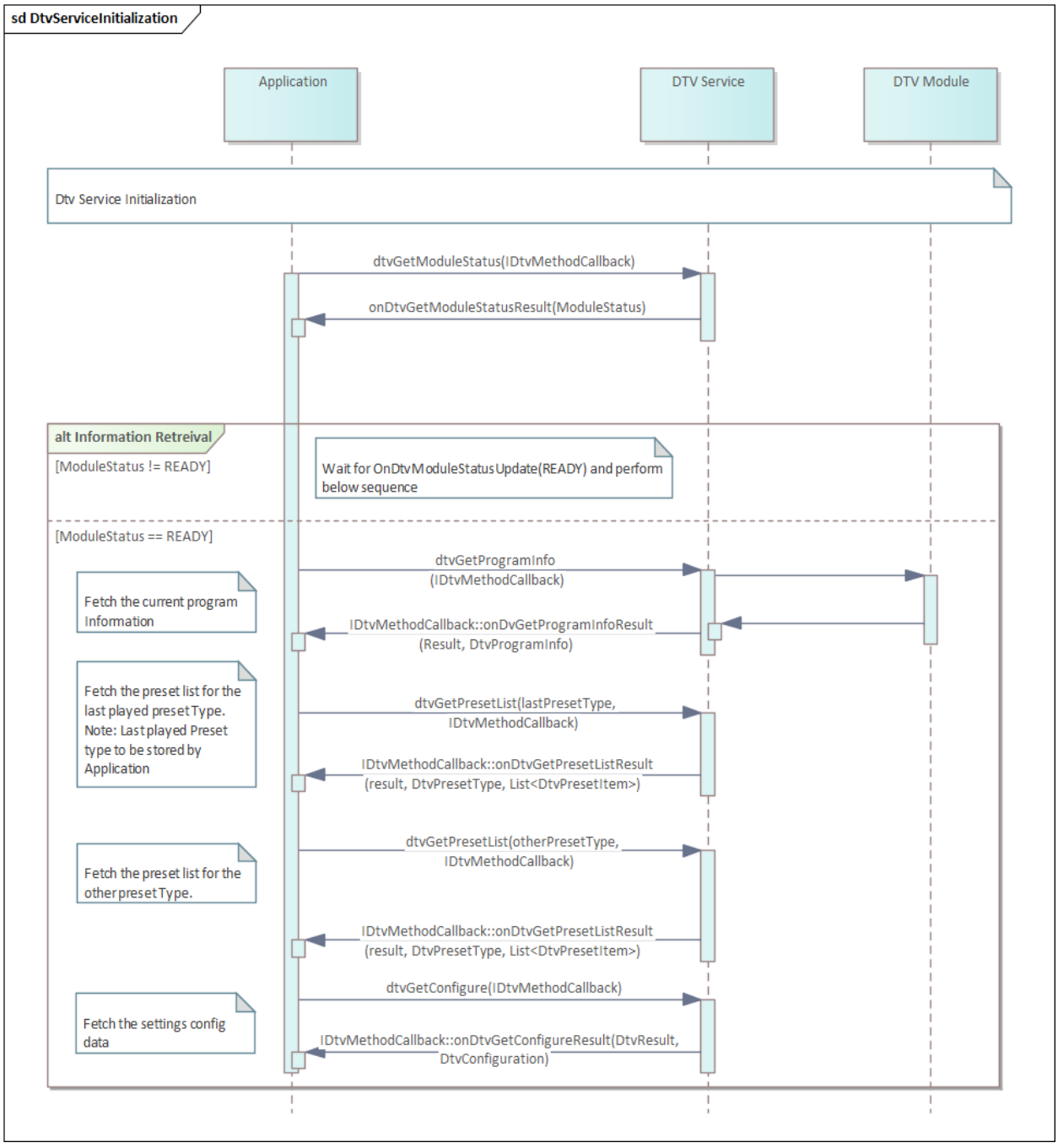
4.1.2 Dtv Service Connected – Handling



4.1.3 Dtv Service Disconnected – Handling

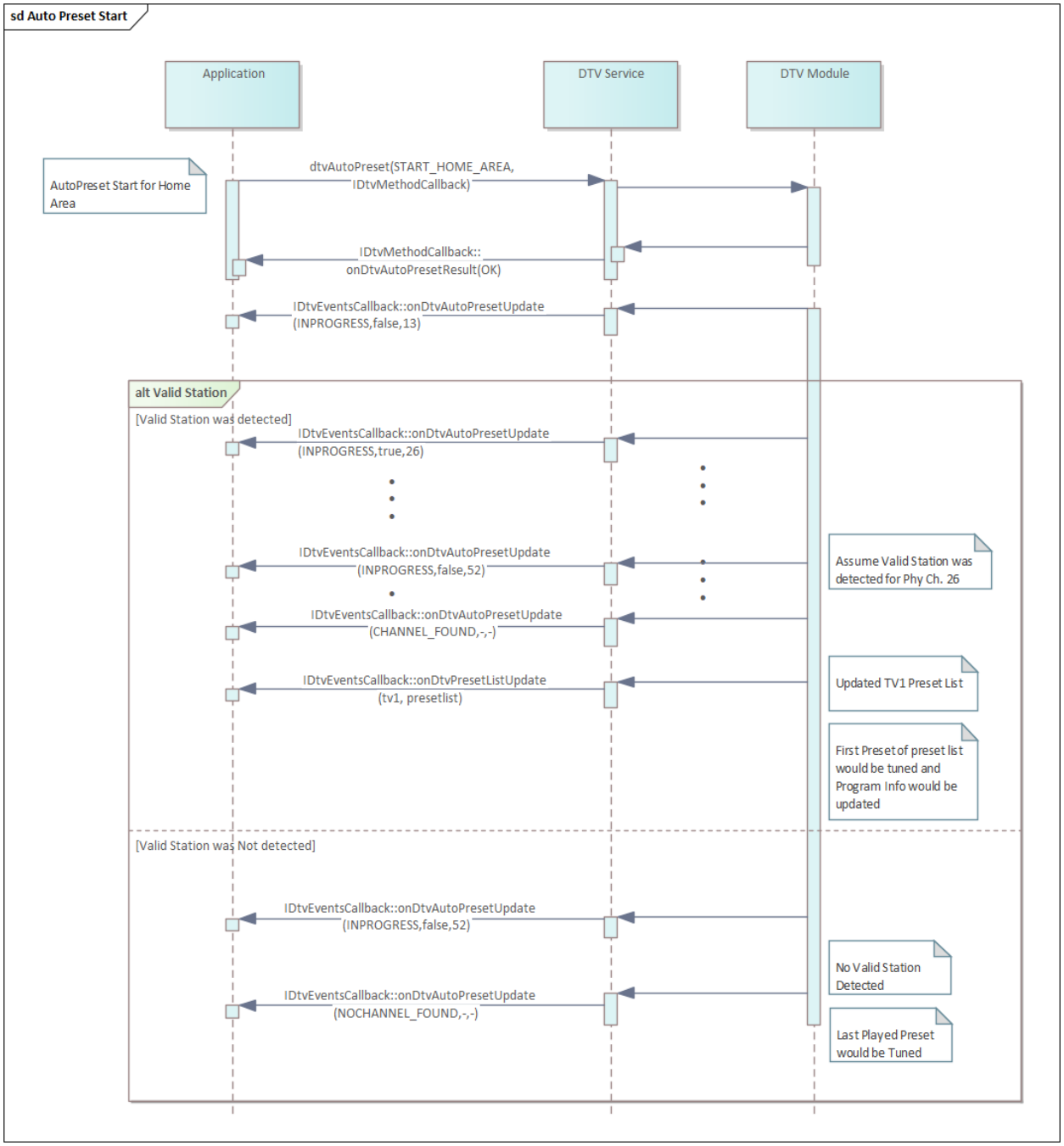


4.1.4 Dtv Service Initialization Sequence

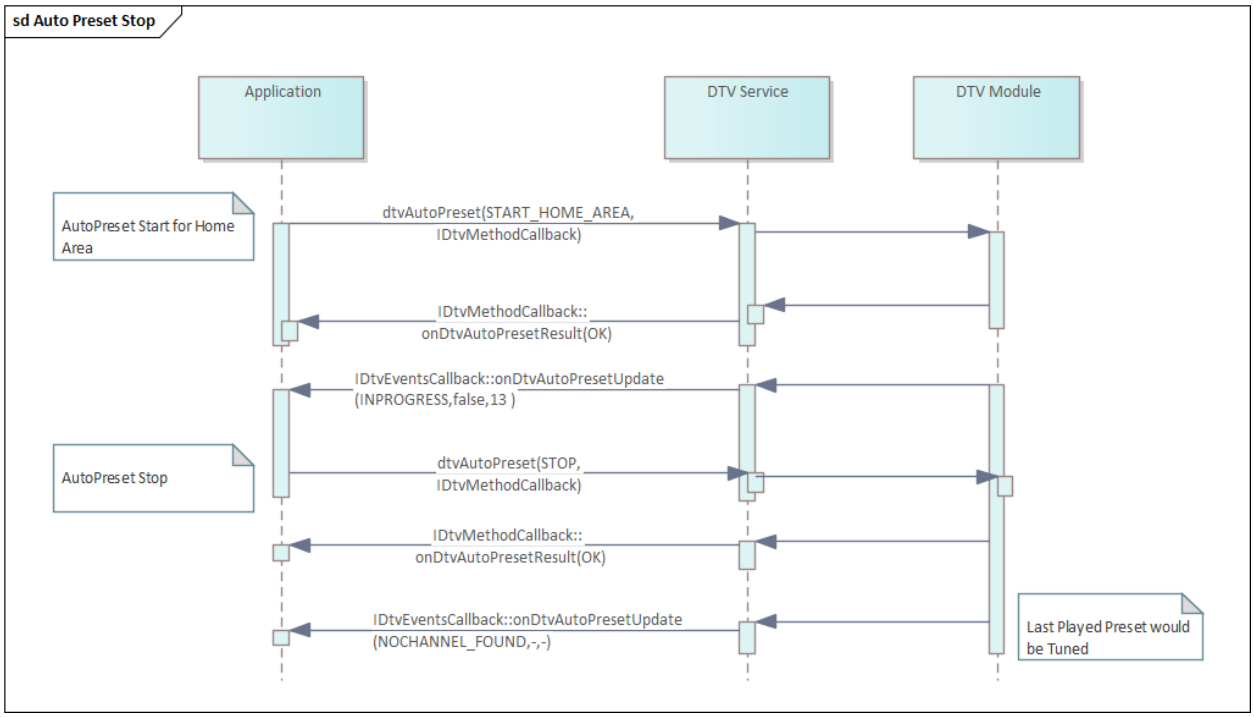


4.2 Auto Preset

4.2.1 Auto Preset Start

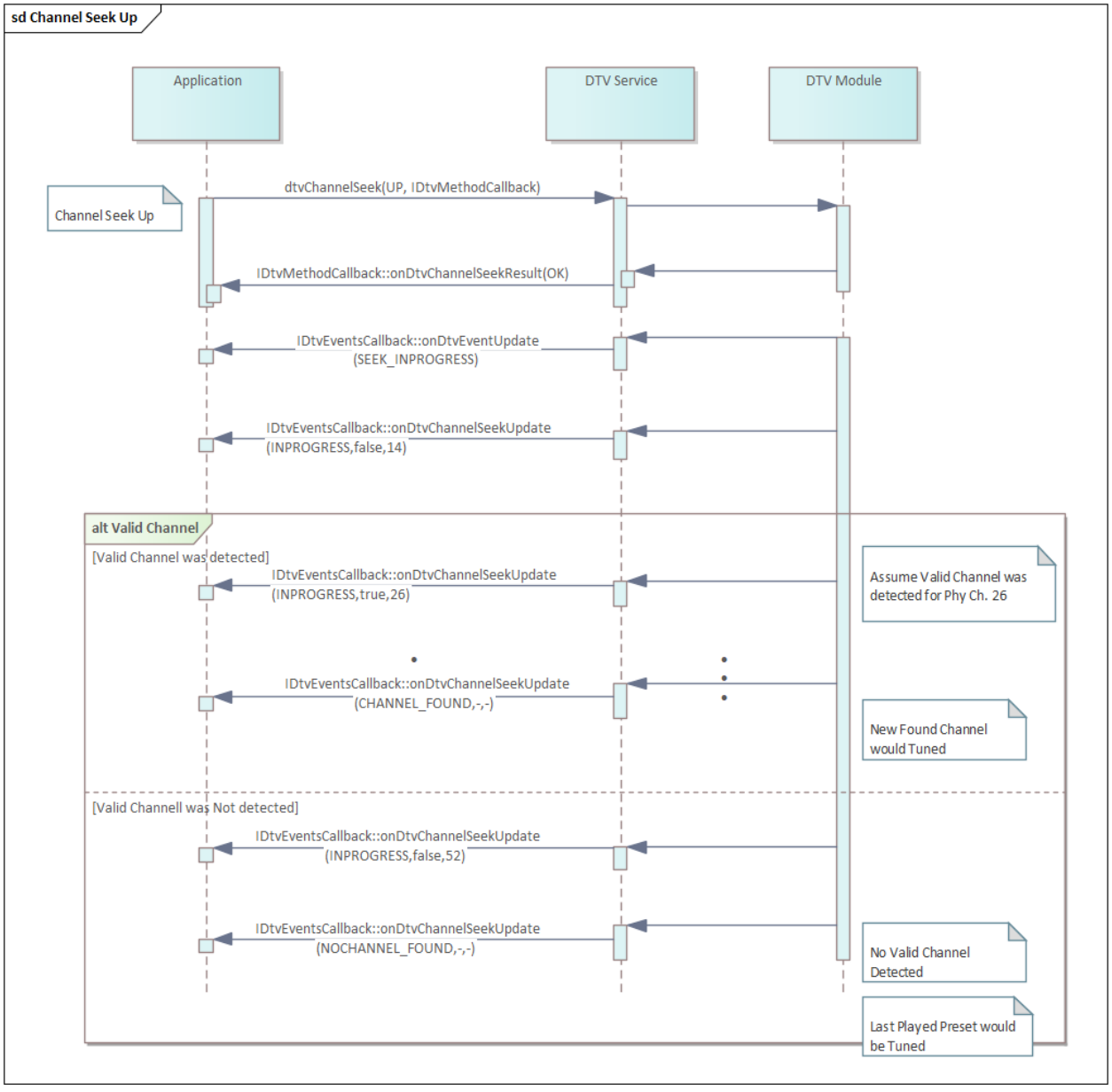


4.2.2 Auto Preset Stop

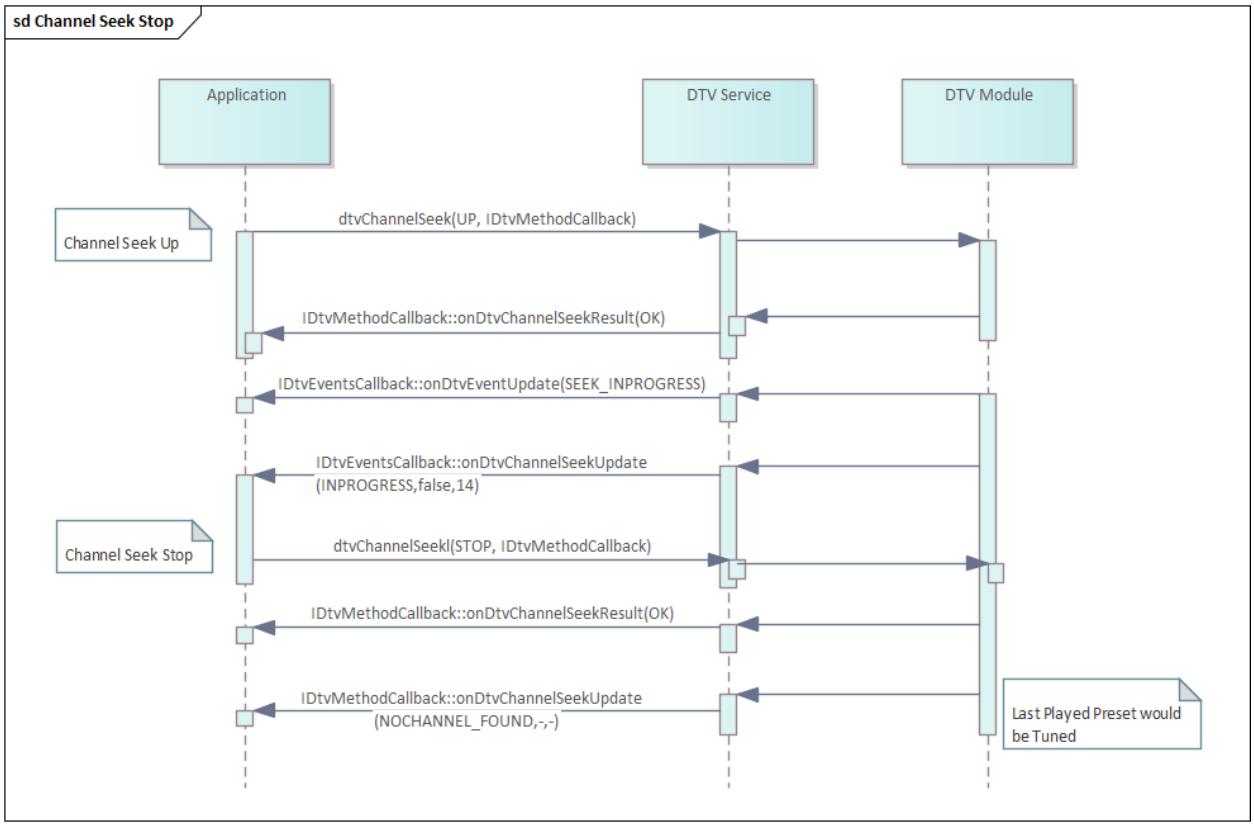


4.3 Channel Seek

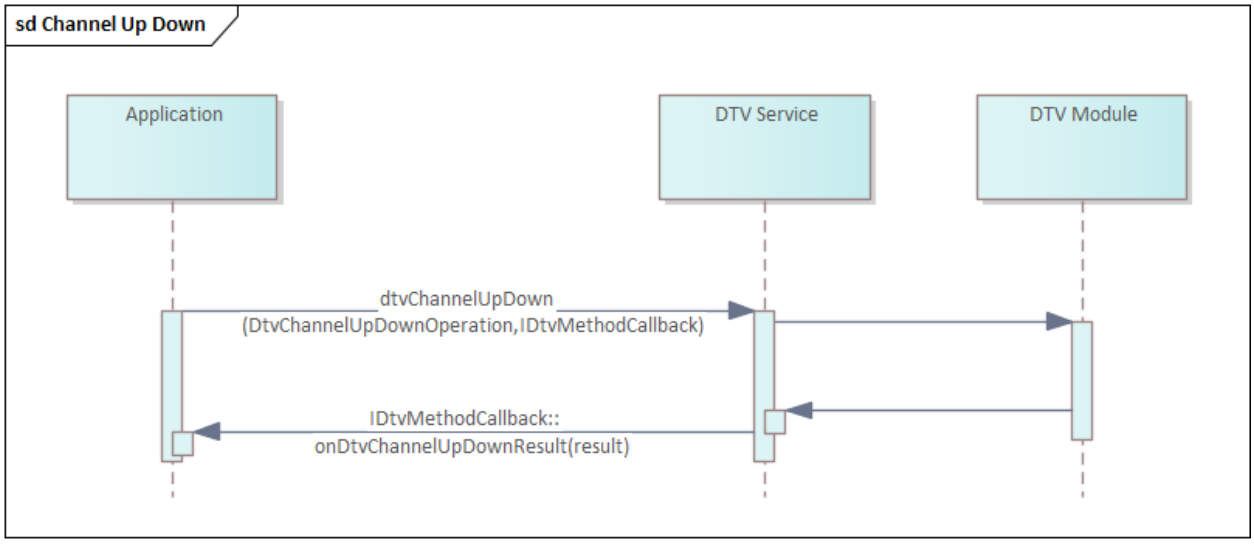
4.3.1 Channel Seek Up



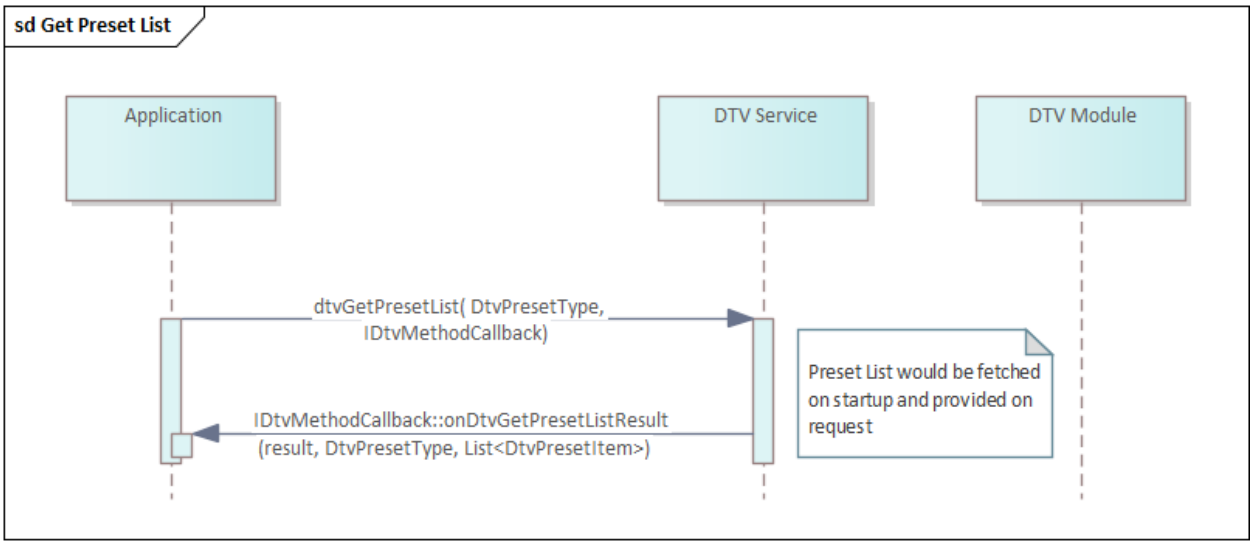
4.3.2 Channel Seek Stop



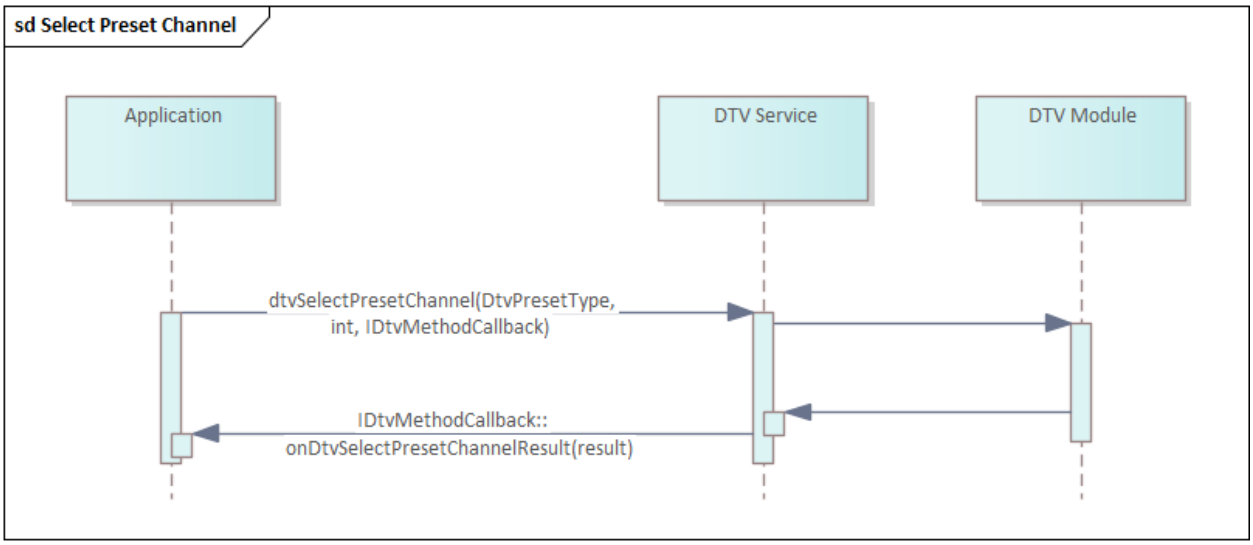
4.4 Channel Up Down



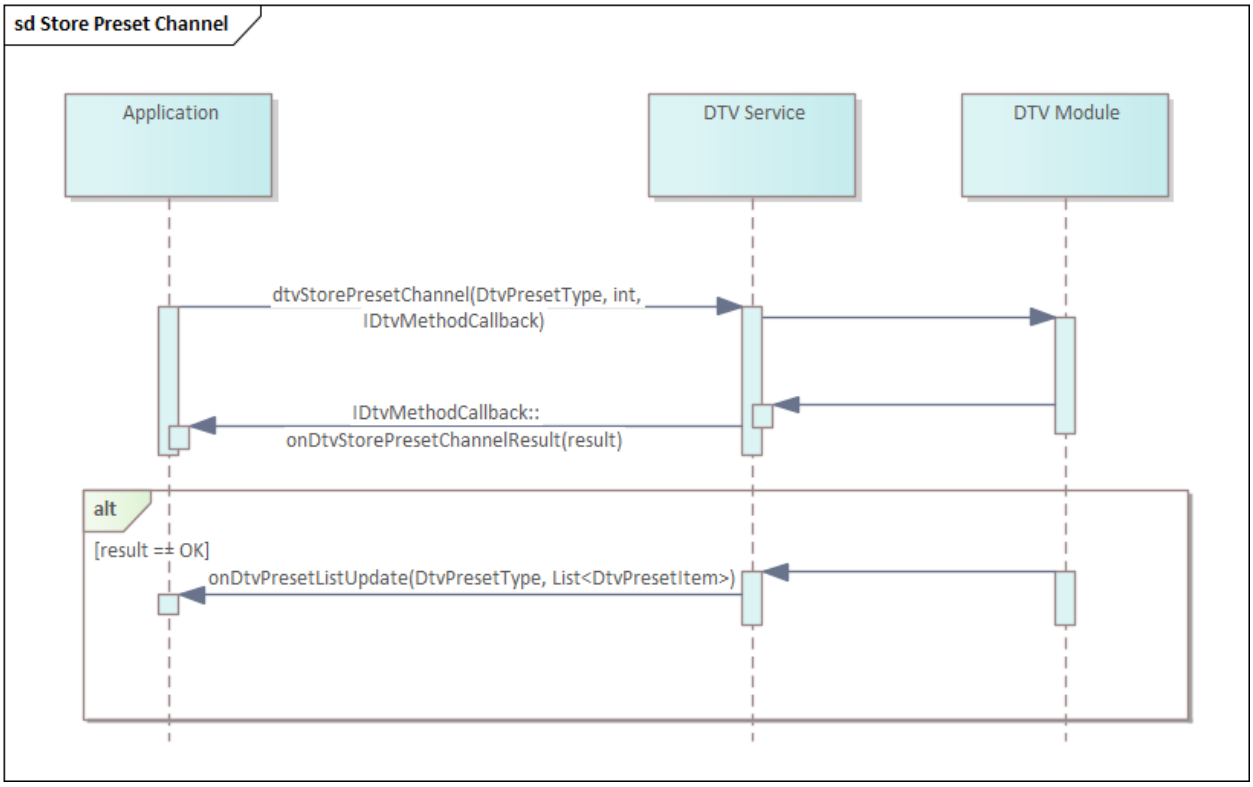
4.5 Get Preset List



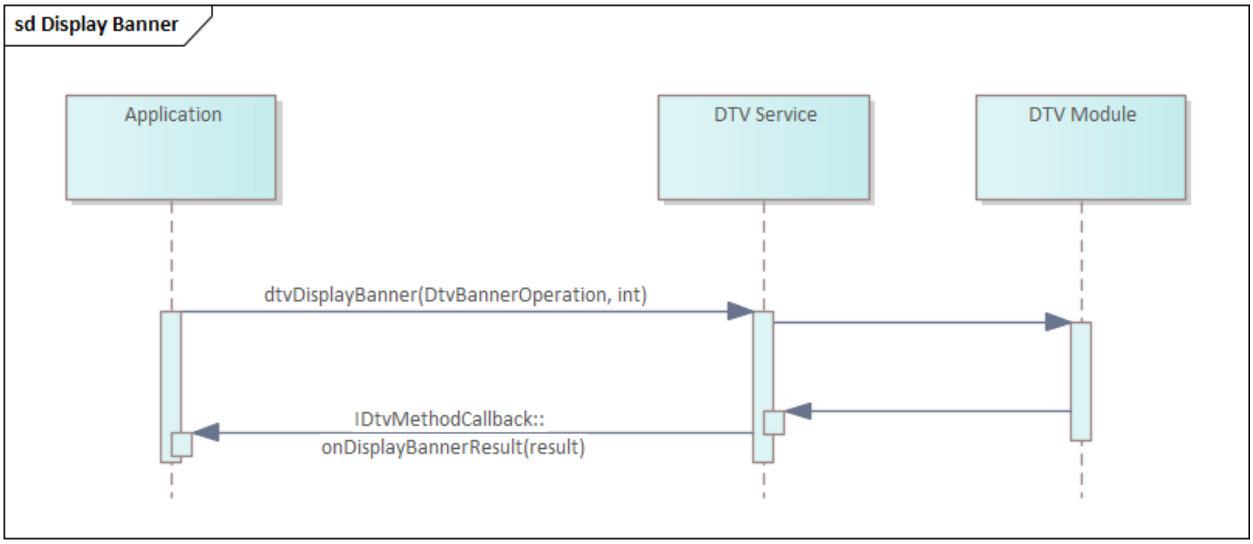
4.6 Select Preset Channel



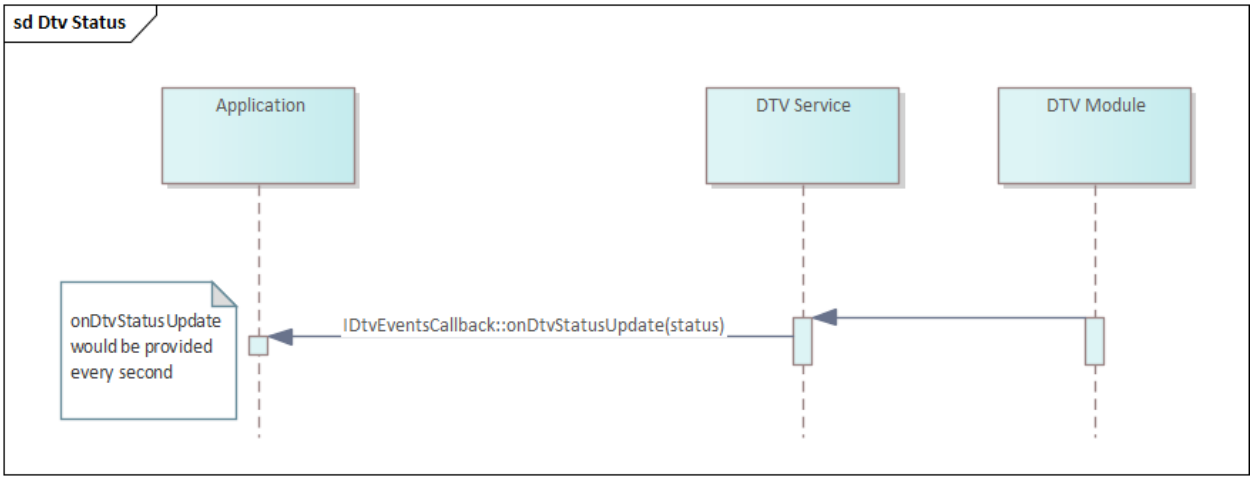
4.7 Store Preset Channel



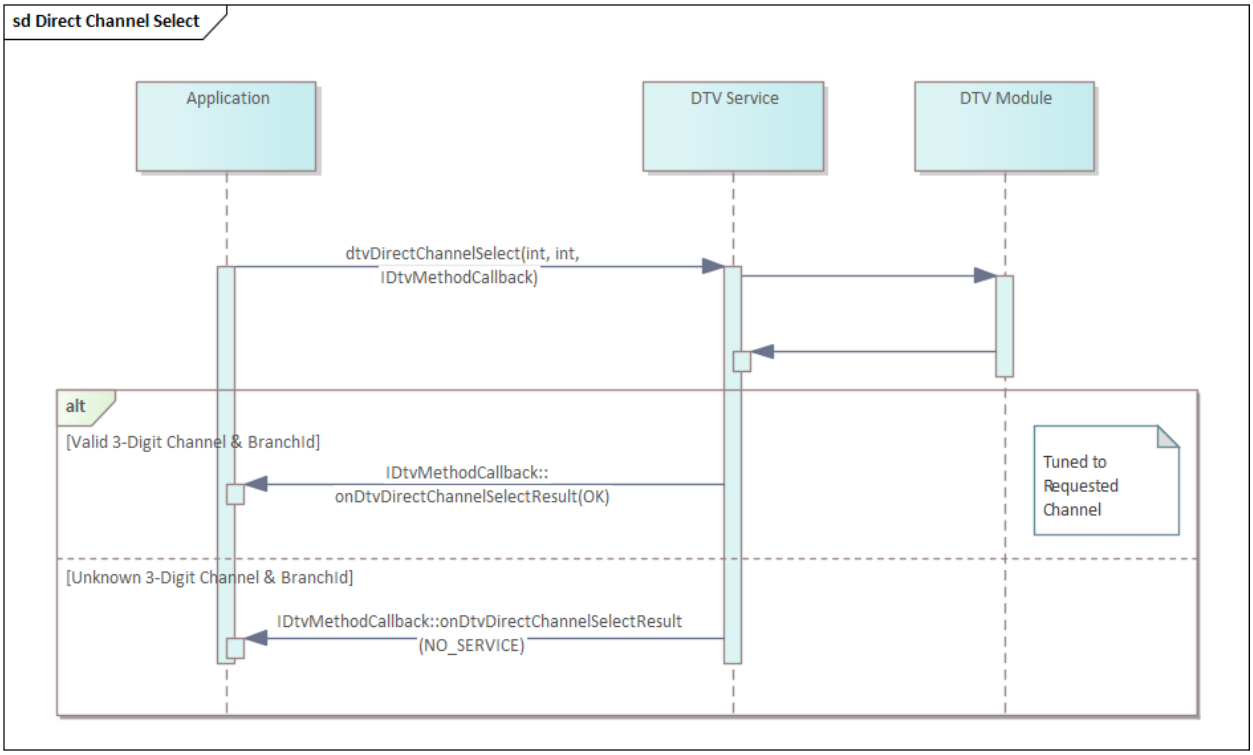
4.8 Display Banner



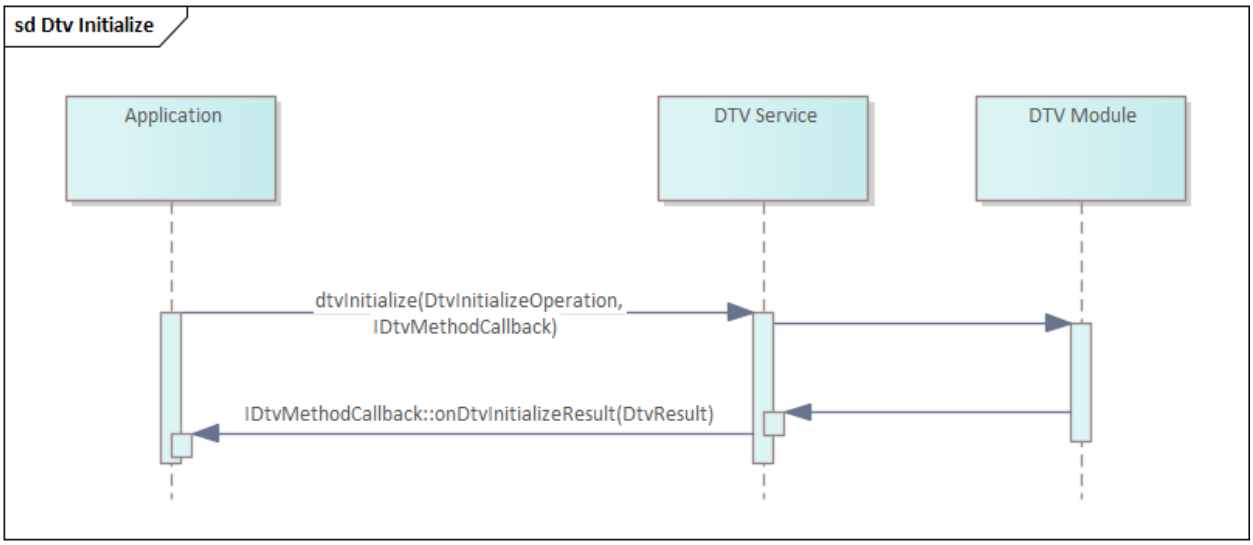
4.9 Dtv Status Update



4.10 Direct Channel Select

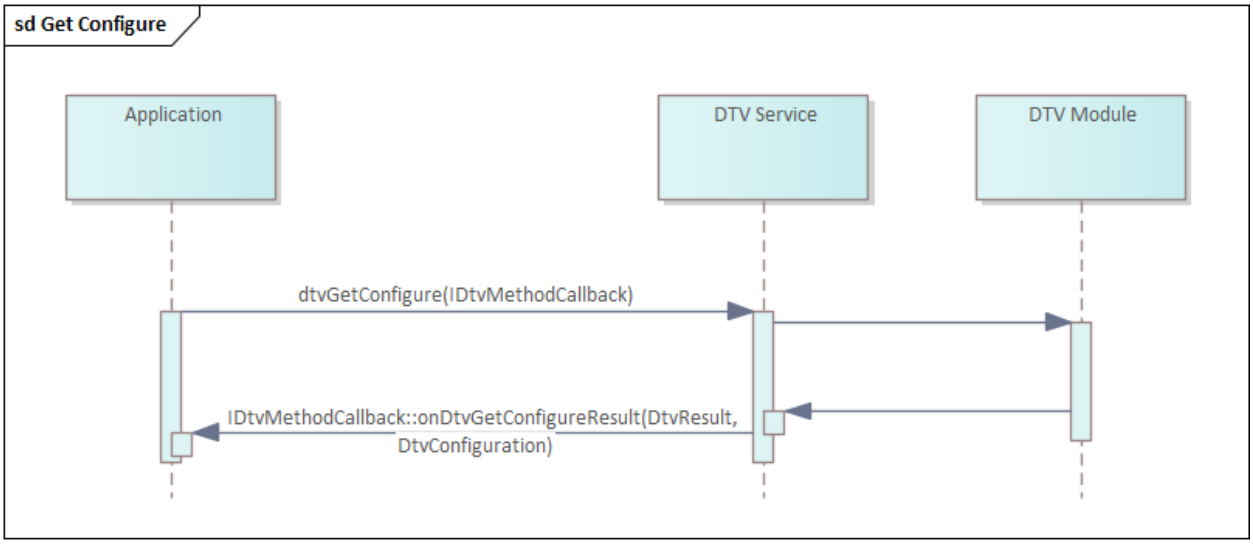


4.11 Dtv Initialize

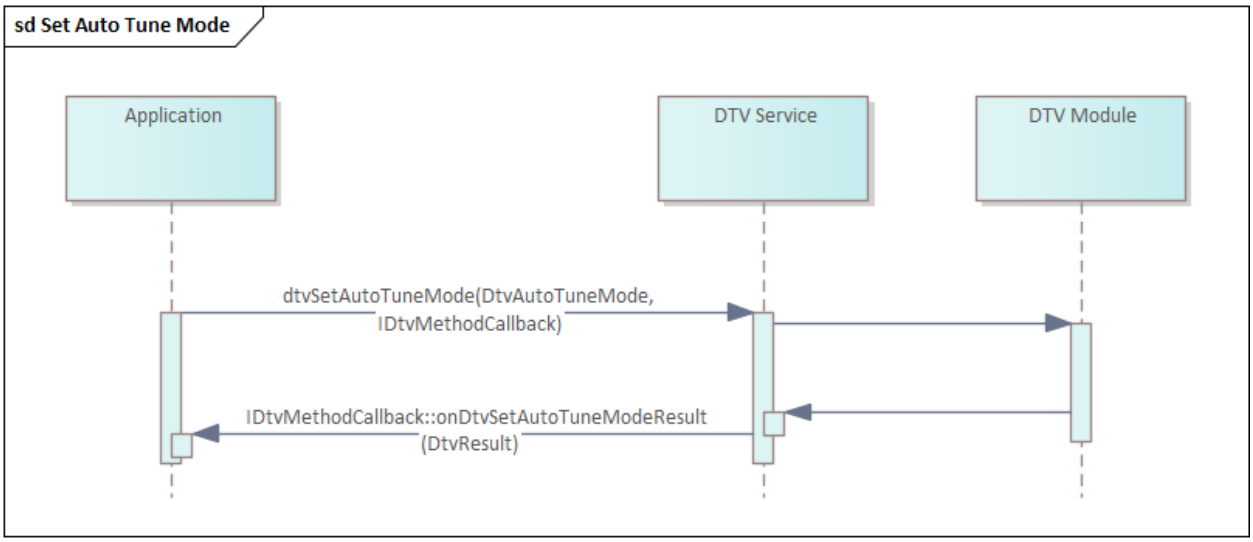


4.12 Dtv Settings Basic

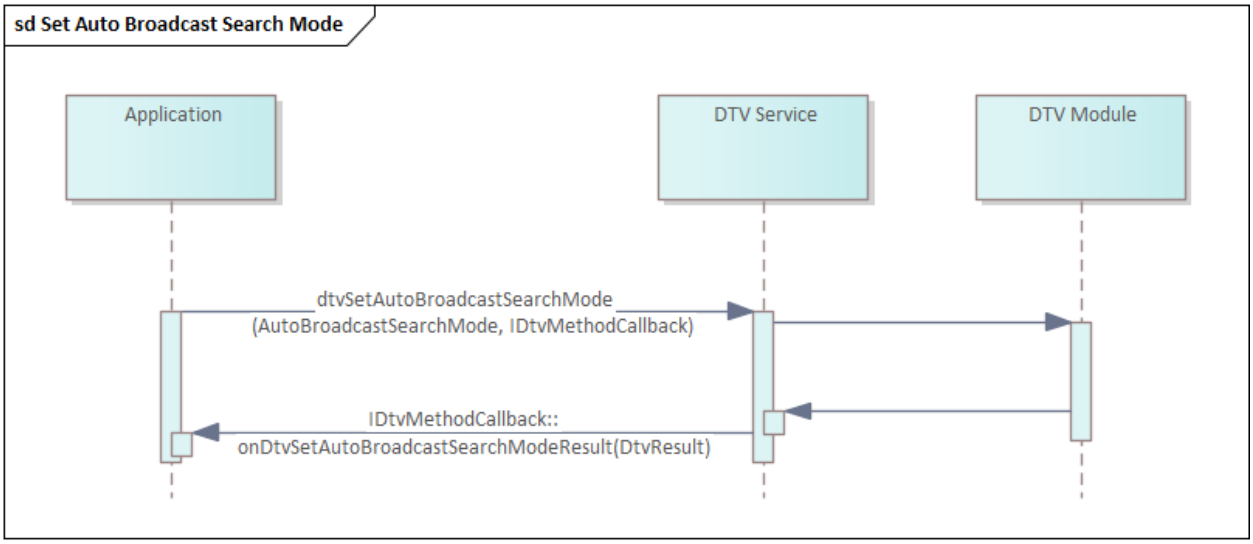
4.12.1 Get Configure



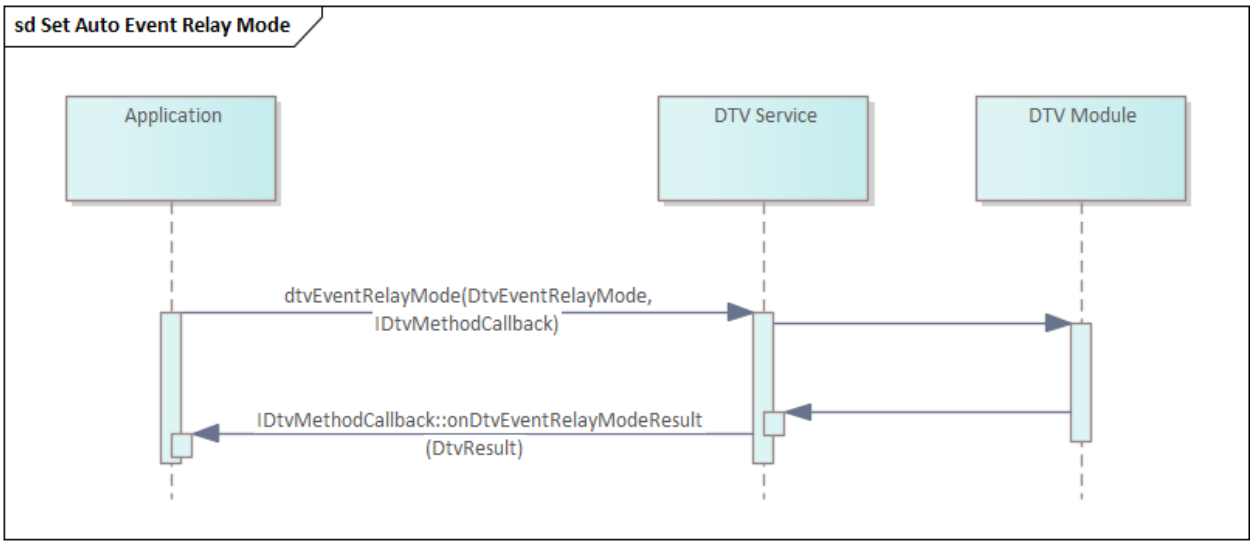
4.12.2 Set Auto Tune Mode



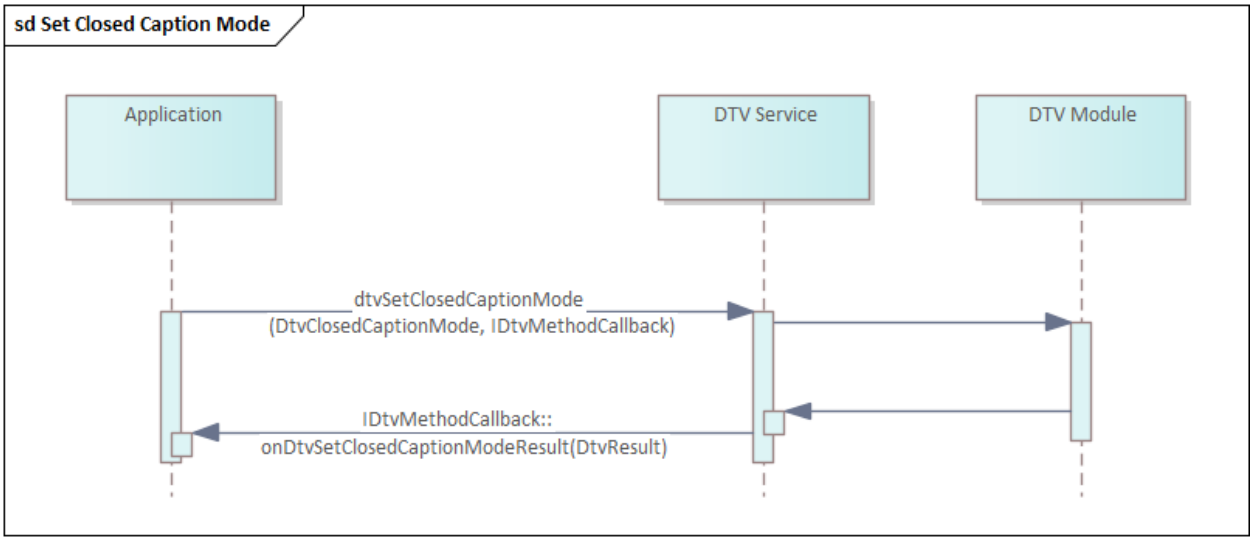
4.12.3 Set Auto Broadcast Search Mode



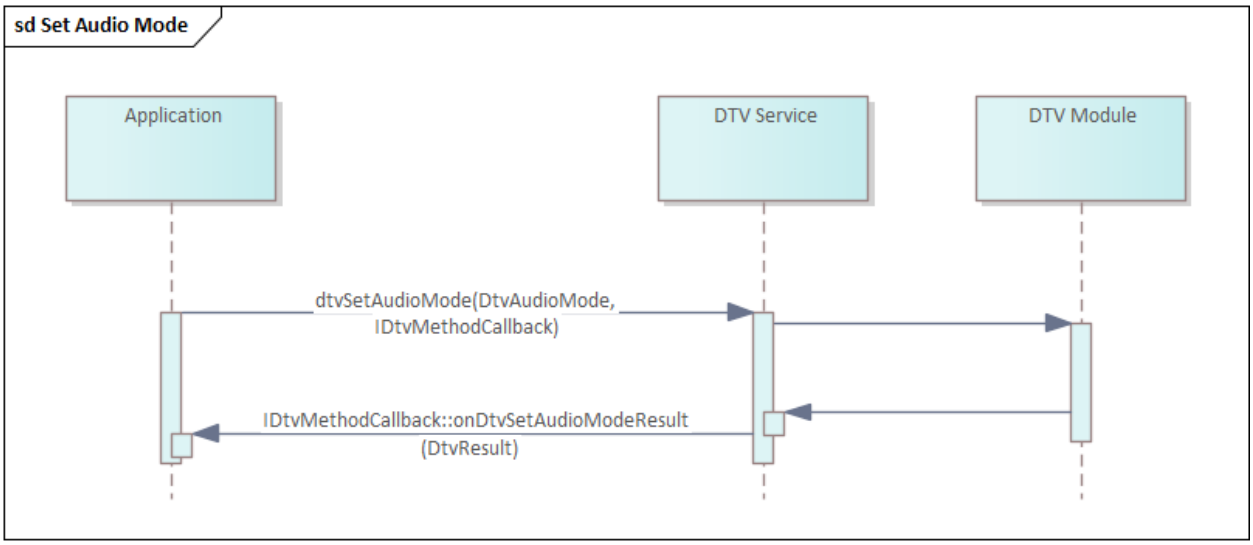
4.12.4 Set Auto Event Relay Mode



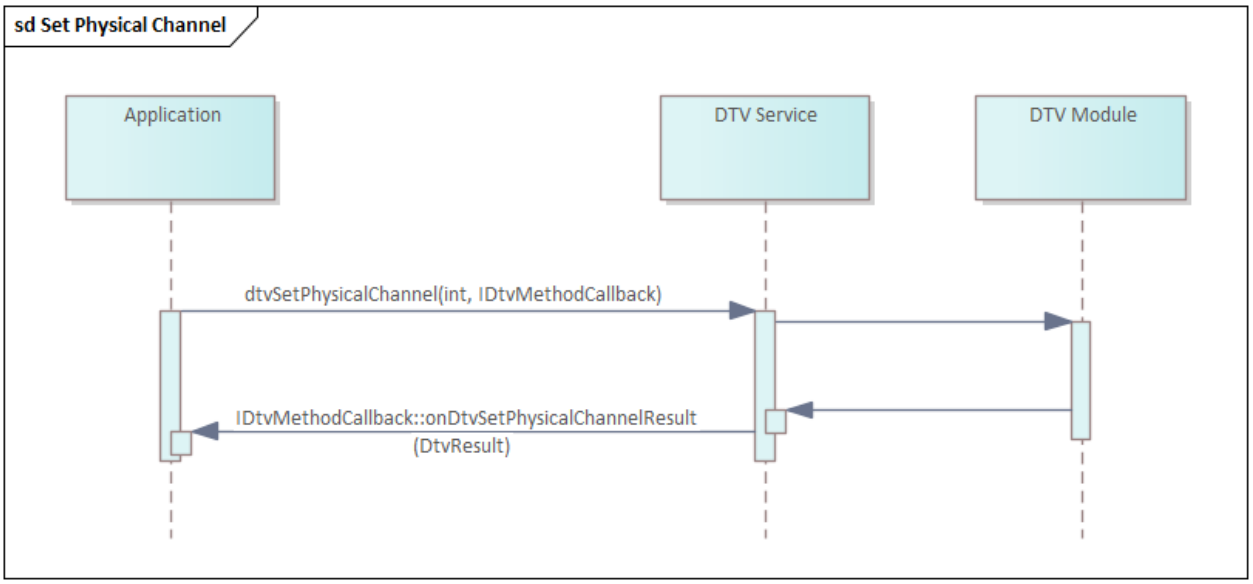
4.12.5 Set Closed Caption Mode



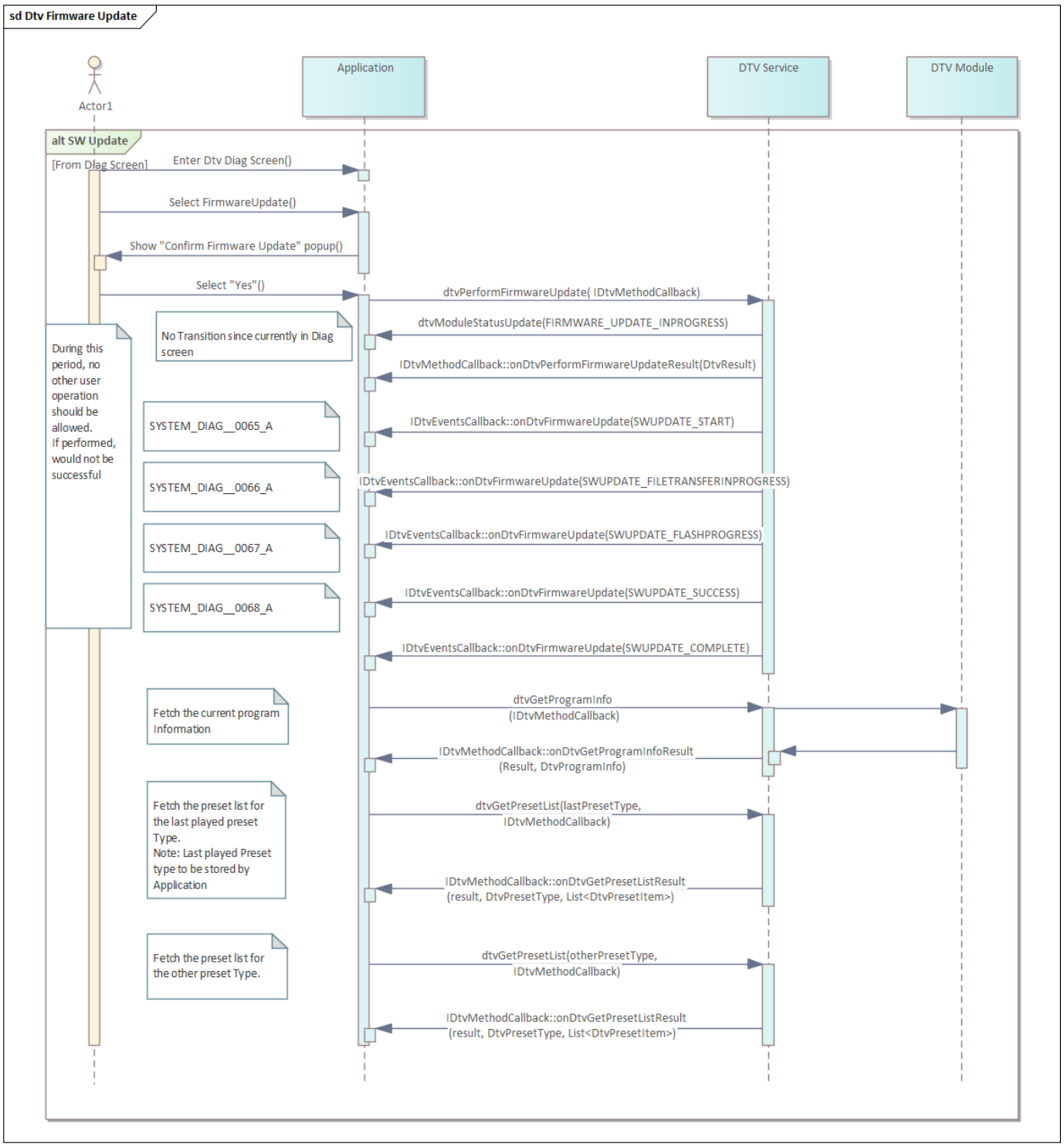
4.12.6 Set Audio Mode



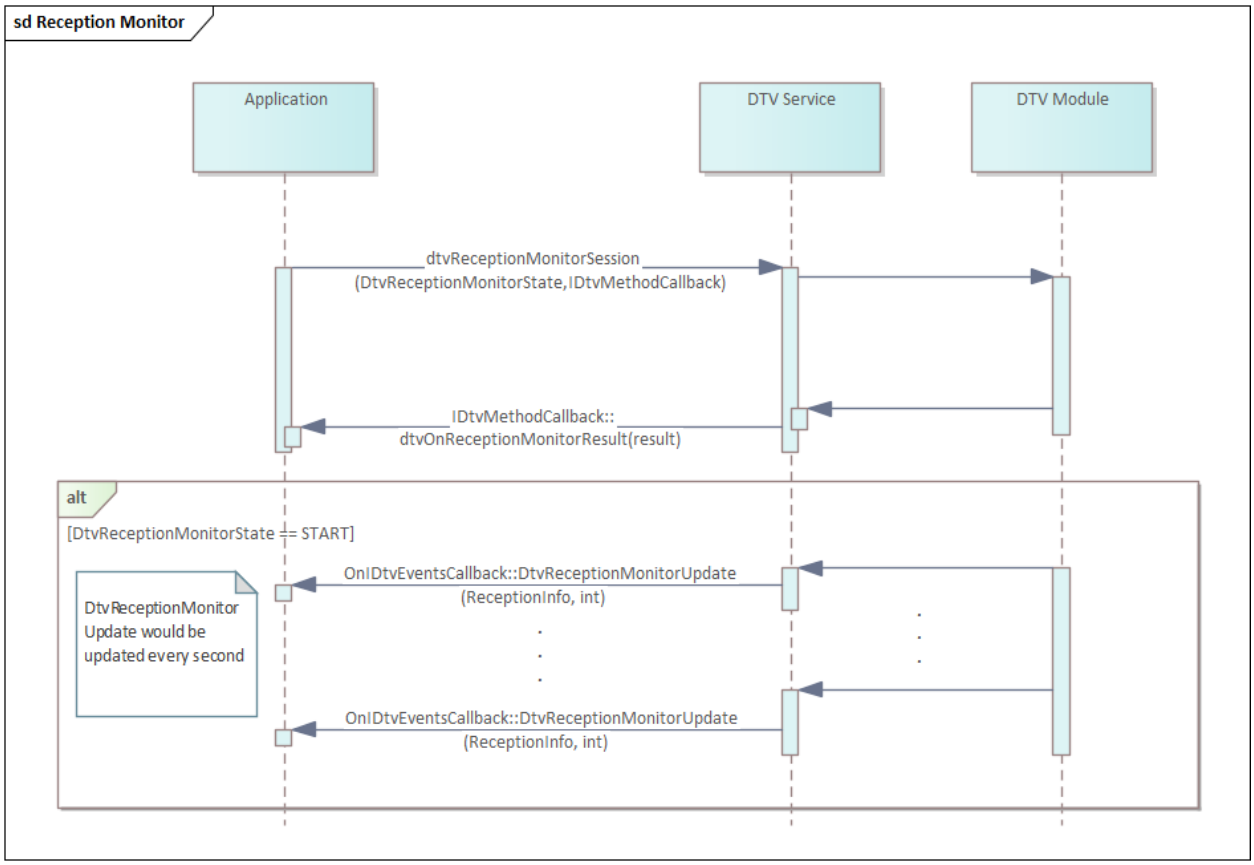
4.13 Set Physical Channel



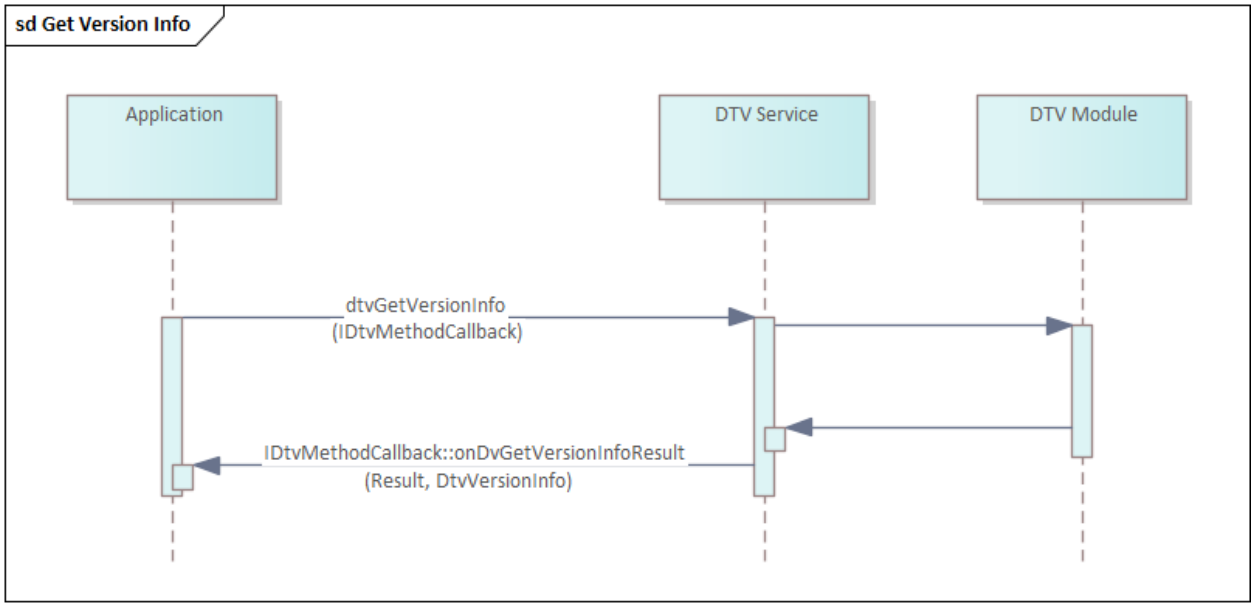
4.14 Dtv Firmware Update



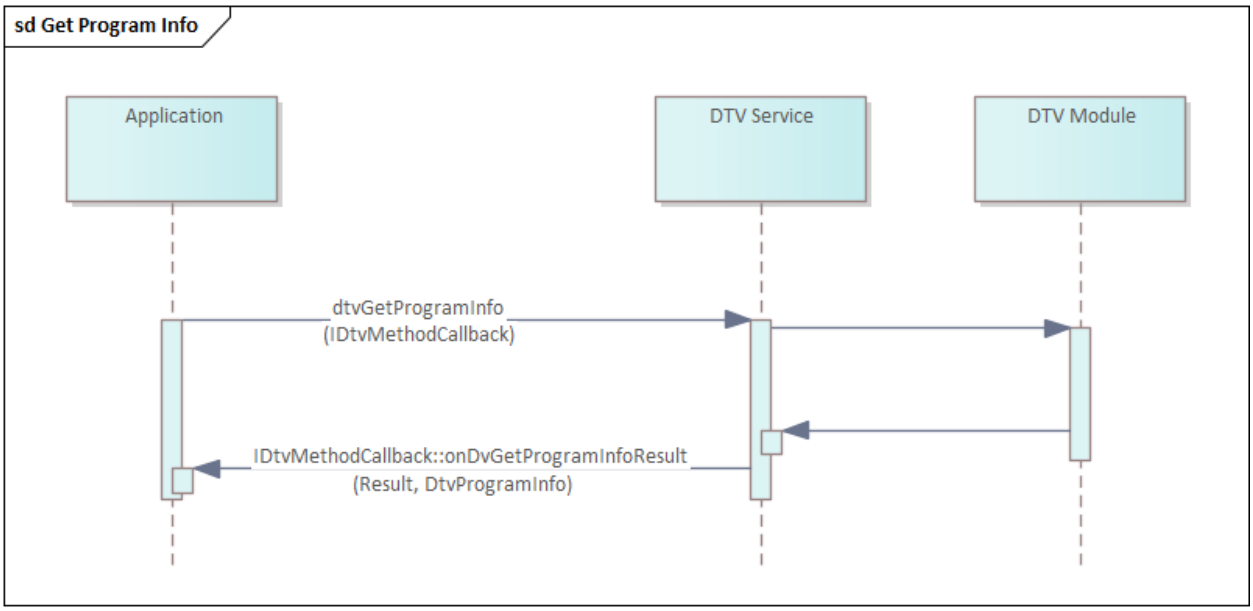
4.15 Reception Monitor (Test Mode)



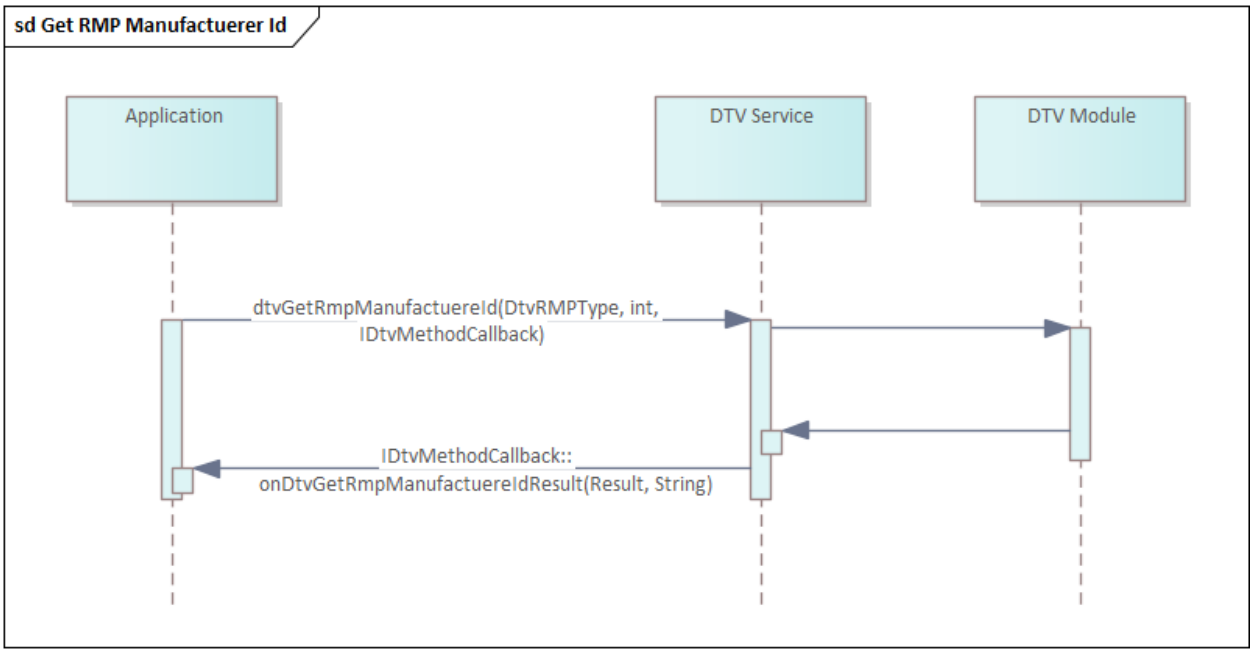
4.16 Get Version Info



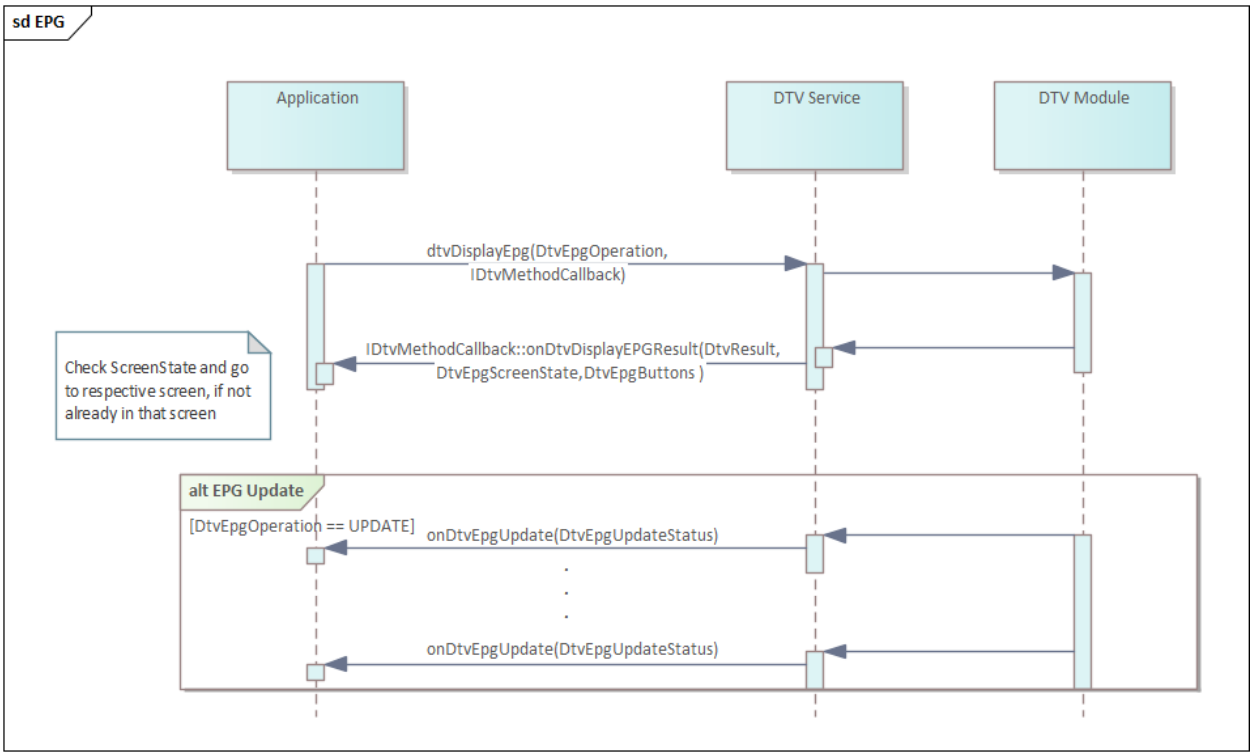
4.17 Get Program Info



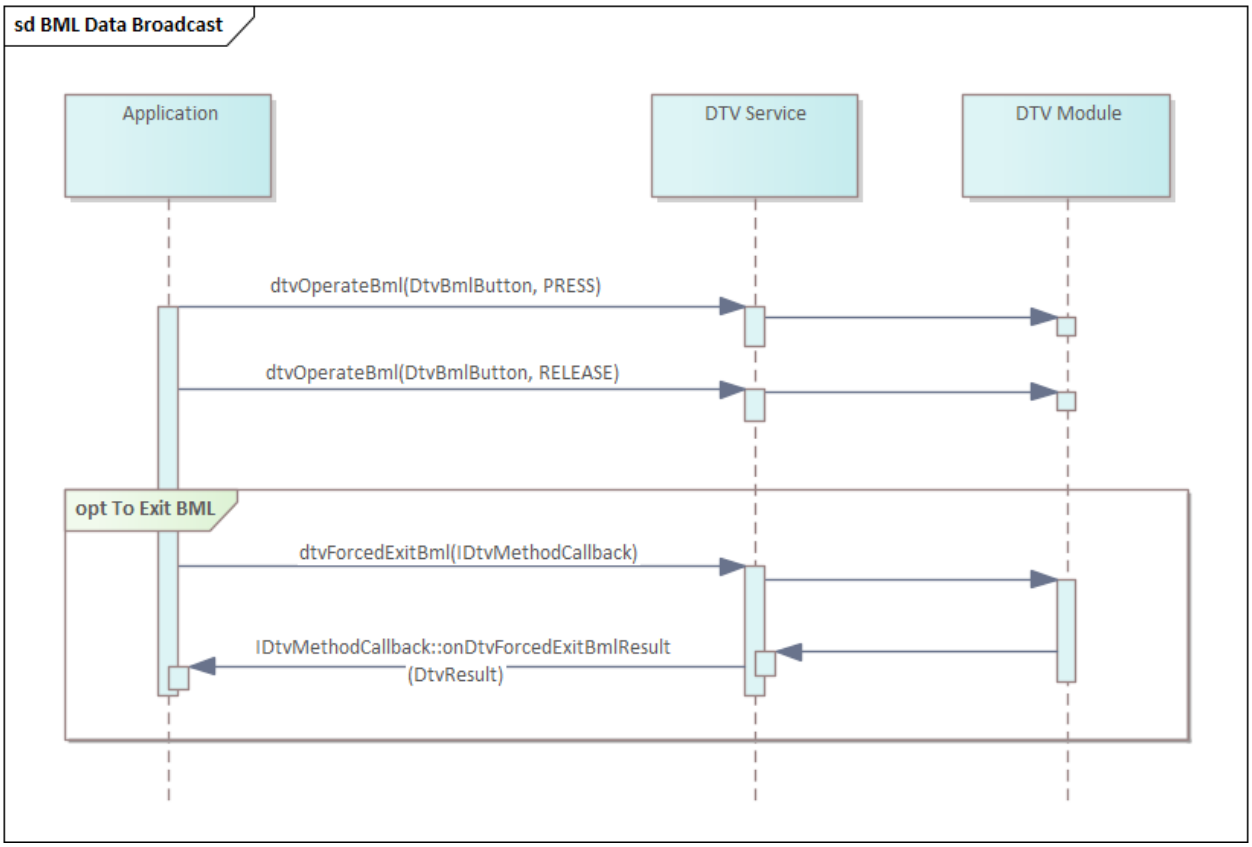
4.18 Get RMP Manufacturer Id



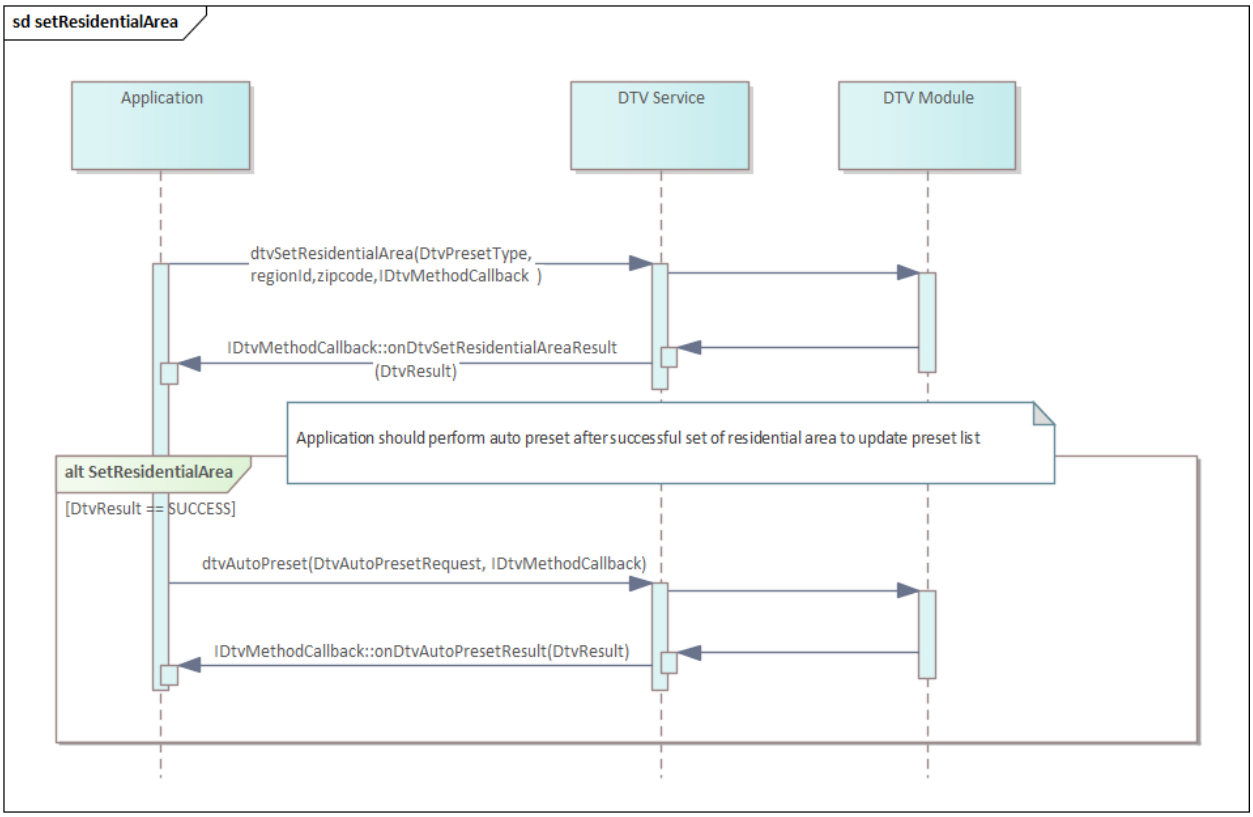
4.19 EPG


sd EPG
Application
DTV Service
DTV Module
dtvDisplayEpg(DtvEpgOperation, IDtvMethodCallback)
IDtvMethodCallback::onDtvDisplayEPGResult(DtvResult, DtvEpgScreenState, DtvEpgButtons)
Check ScreenState and go to respective screen, if not already in that screen
alt EPG Update
[DtvEpgOperation == UPDATE]
onDtvEpgUpdate(DtvEpgUpdateStatus)
onDtvEpgUpdate(DtvEpgUpdateStatus)

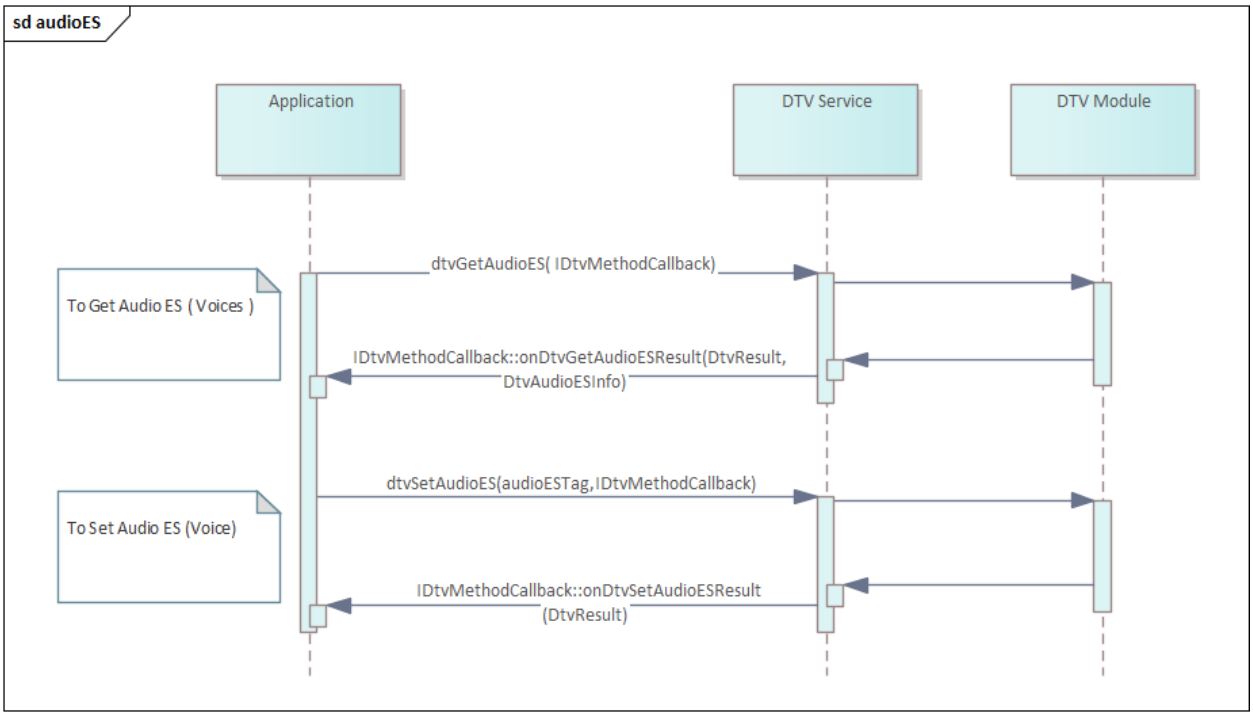
4.20 BML Data Broadcast



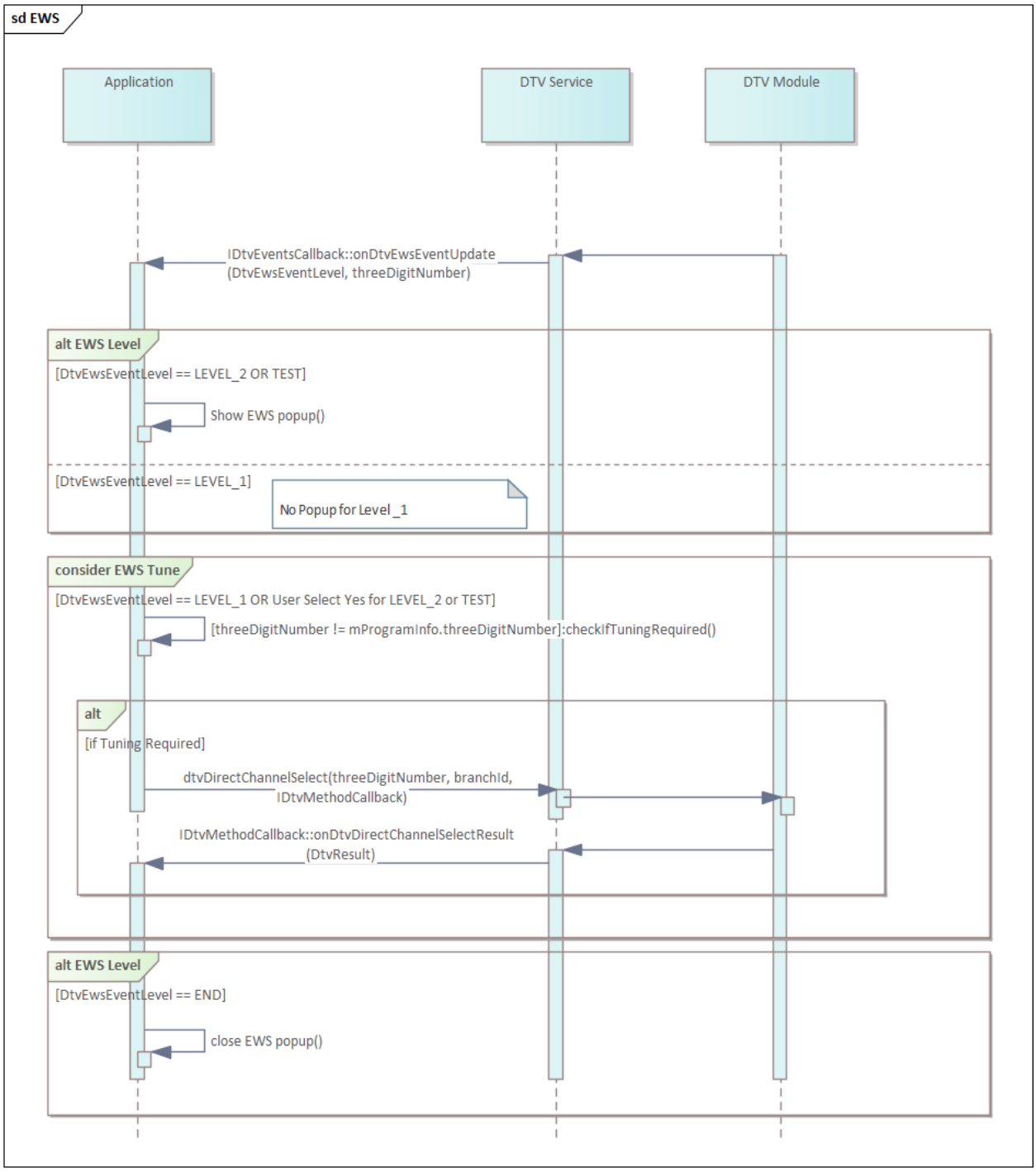
4.21 Set Residential Area



4.22 Audio ES

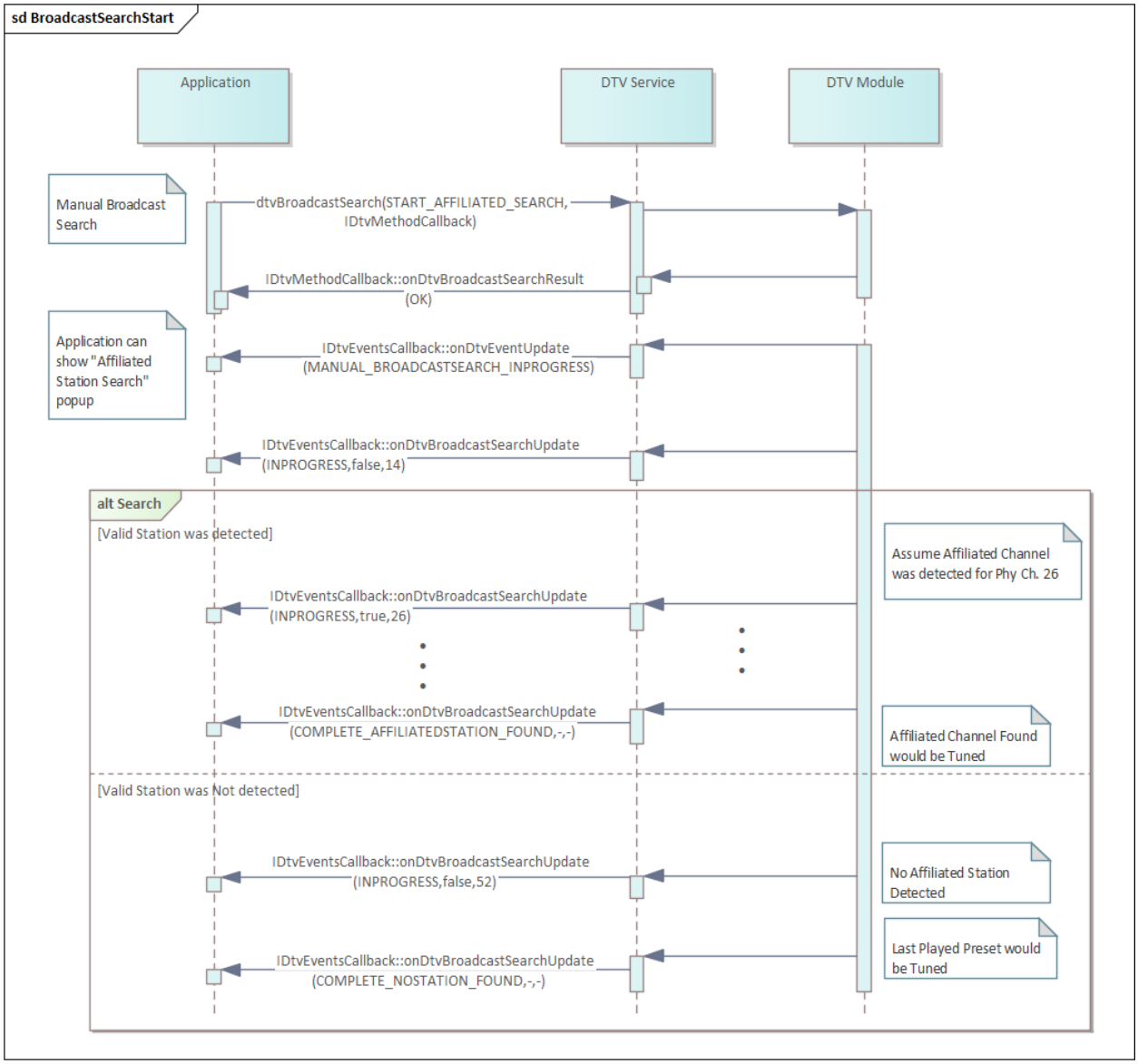

sd audioES
Application
DTV Service
DTV Module
dtvGetAudioES(IDtvMethodCallback)
To Get Audio ES (Voices)
IDtvMethodCallback::onDtvGetAudioESResult(DtvResult, DtvAudioESInfo)
dtvSetAudioES(audioESTag,IDtvMethodCallback)
To Set Audio ES (Voice)
IDtvMethodCallback::onDtvSetAudioESResult (DtvResult)

4.23 EWS

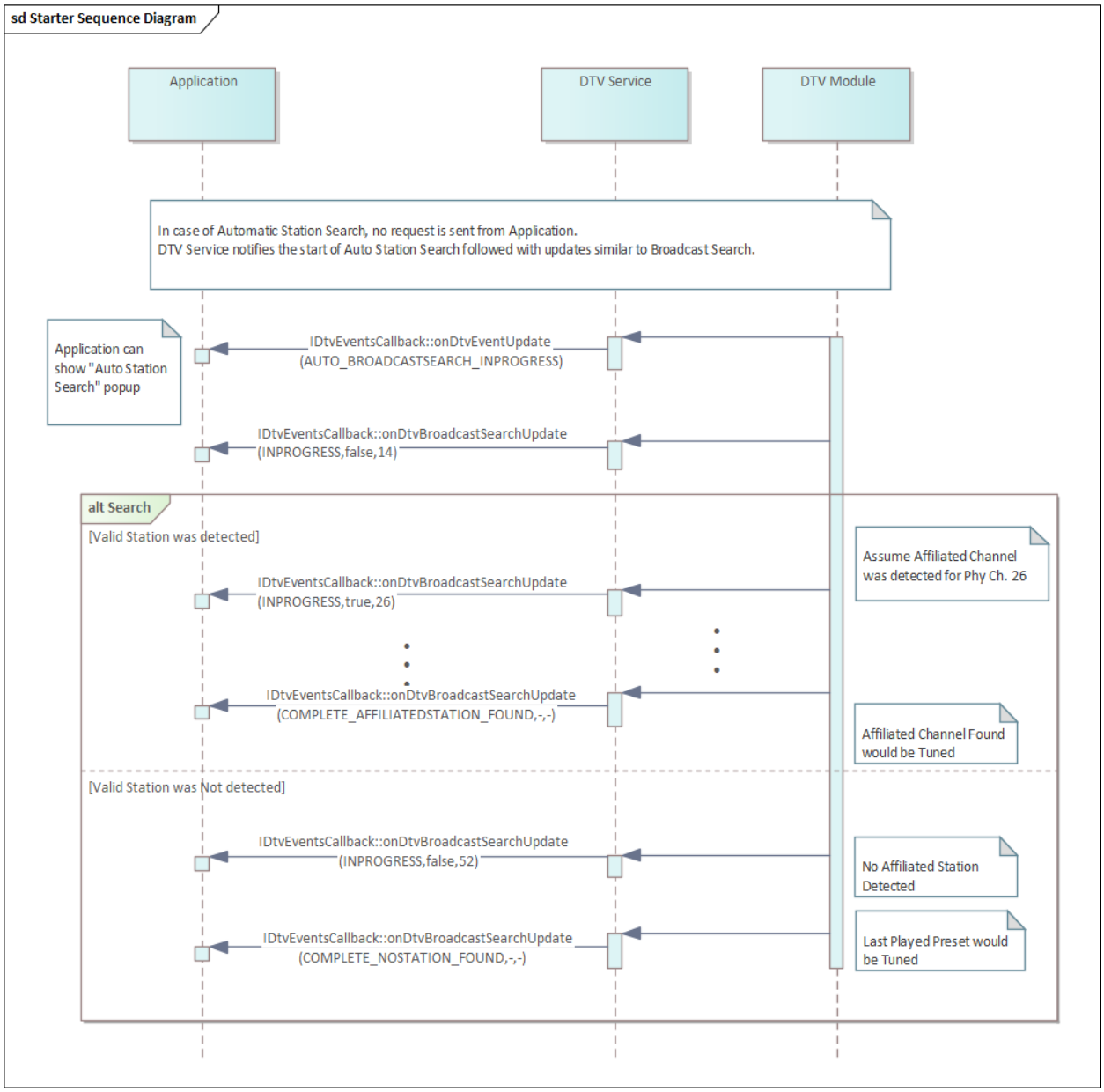

sd EWS
Application
DTV Service
DTV Module
IDtvEventsCallback::onDtvEwsEventUpdate
(DtvEwsEventLevel, threeDigitNumber)
alt EWS Level
[DtvEwsEventLevel == LEVEL_2 OR TEST]
Show EWS popup()
[DtvEwsEventLevel == LEVEL_1]
No Popup for Level _1
consider EWS Tune
[DtvEwsEventLevel == LEVEL_1 OR User Select Yes for LEVEL_2 or TEST]
[threeDigitNumber != mProgramInfo.threeDigitNumber]:checkIfTuningRequired()
alt
[if Tuning Required]
dtvDirectChannelSelect(threeDigitNumber, branchId,
IDtvMethodCallback)
IDtvMethodCallback::onDtvDirectChannelSelectResult
(DtvResult)
alt EWS Level
[DtvEwsEventLevel == END]
close EWS popup()

4.24 Broadcast Search

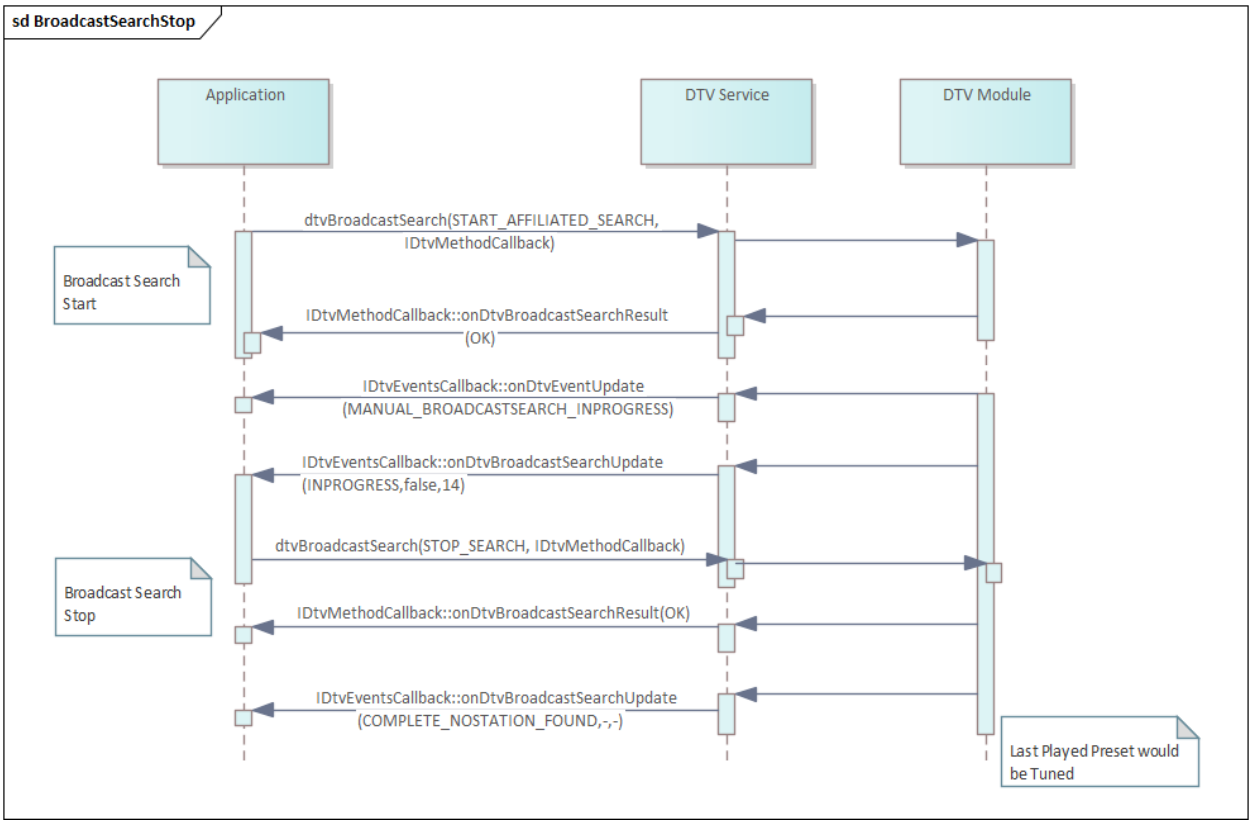
4.24.1 Manual Broadcast Search (Affilaited Search)



4.24.2 Auto Station Search

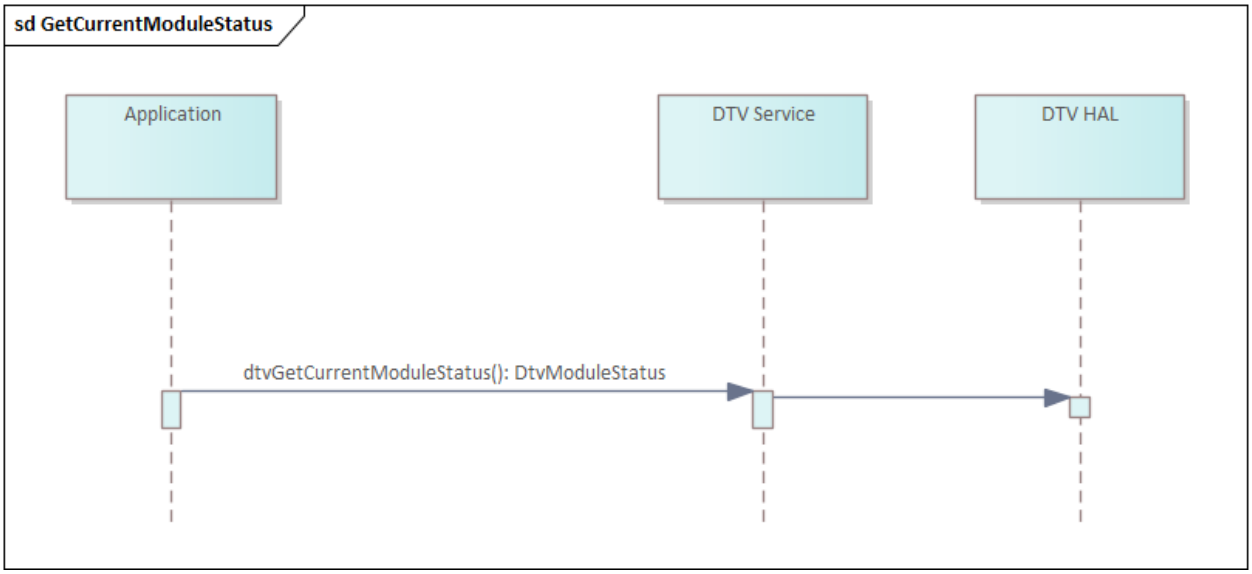


4.24.3 Broadcast Search Stop (Affiliated Search and Auto Station Search)

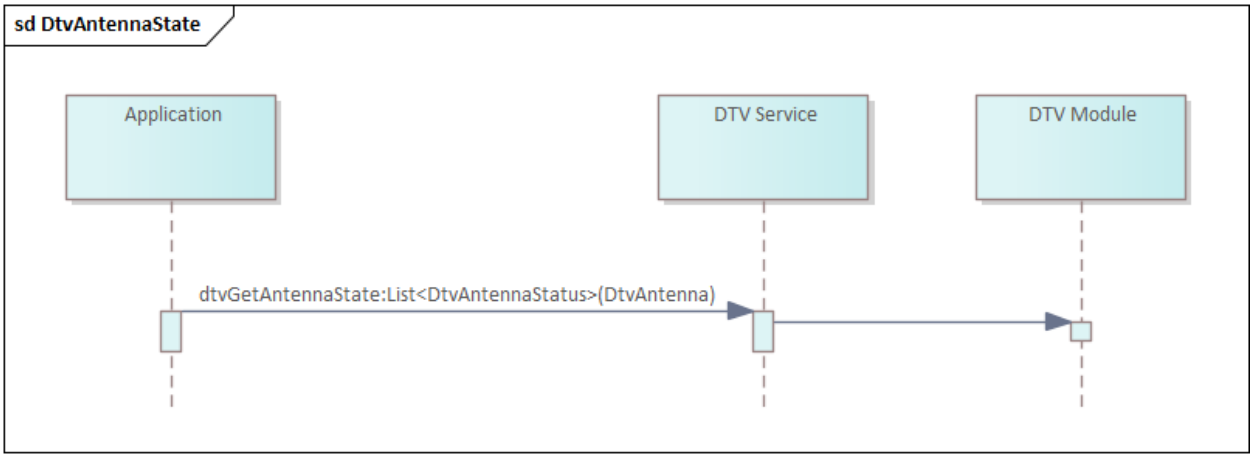


4.25 Dtv Module Status

4.25.1 Get Current Module Status

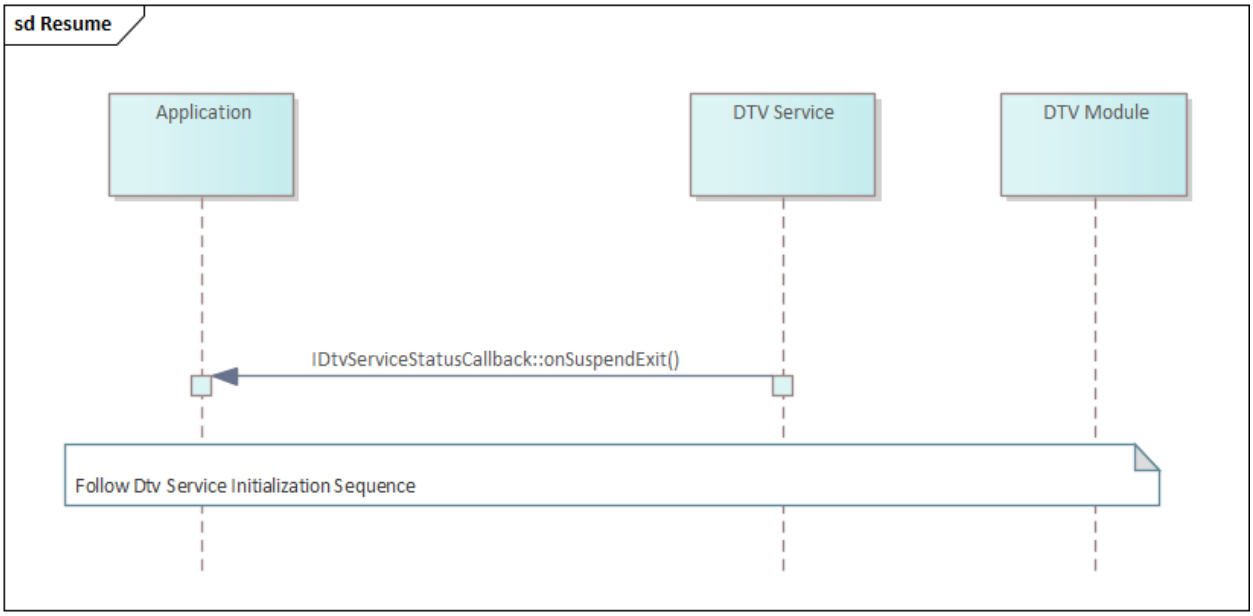


4.26 Dtv Antenna State



4.27 Suspend to Disk

4.27.1 Resume



4.27.2 Suspend

